



Myndigheten för
samhällsskydd
och beredskap

GUIDANCE

Guidance in forest fire extinguishing

3rd edition 2022



Forest Fire Fighting Guide 3rd Edition

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Preface

Over the past few years, we have experienced several extensive forest fires in Sweden. Climate change, with drier conditions, means that the likelihood of large and difficult-to-extinguish forest fires will increase over time.

Therefore, there is a need to increase knowledge about forest fire fighting so that we are better prepared for upcoming forest fire seasons. The changing risk picture will also increase the need for methods that are adapted to the new conditions.

This guidance focuses on forest fire control, both on the ground and from the air.

The target group for the guidance is primarily municipal rescue services, and above all personnel who actively participate in firefighting in forests and land.

Consider this guidance as a living document that will be updated over time.

Cecilia Loostrom

Department Head

The Swedish Civil Contingencies Agency

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Introduction

1. Introduction

1.1 Purpose of the guidance

The Swedish Civil Contingencies Agency (MSB) has, together with a number of municipal rescue services, developed this guidance to

- strengthen knowledge about the practical measures and considerations needed during a forest fire response
- strengthen knowledge about the use of flying resources with the aim of optimizing the use of resources
- strengthen knowledge about fire behavior, especially in extreme conditions
- strengthen knowledge about risk reduction measures
- provide experience and knowledge transfer from previous forest fires occurred.

Areas not covered in this version will be included in future versions. The guidance is therefore not a comprehensive document regarding forest fire extinguishing.

Chapter 16 of this guidance (tactics for extinguishing forest fires) has a summarizing role and is intended to show a common thread for the various components of the guidance.

Staff and management methodology for forest fires is not covered in this guidance.

1.2 Existing knowledge material

Existing knowledge material on forest fire extinguishing is primarily found in two places:

- MSB's web-based training on forest fires at fortbildning.msb.se
- the textbook *Forest Fire Fighting* (SRV, 2003).

In particular, the textbook *Forest Fire Fighting* is a number of years old and is in need of revision, additions, etc. The areas covered in this guide are therefore intended to replace the corresponding material in the textbook. Other material in the textbook such as basic fire behavior, weather, etc. remains valid until further notice.

1.3 Supplement to the 3rd edition 2022

In this edition, the following sections or areas have been added:

- Forest fire detection • Fire risk forecasts • Unmanned aerial vehicles

Revisions and additions have been made in the following sections or areas:

- Fire whirls and aerial fires • Extinguishing agent additives
- MSB reinforcement resources • Request for airborne support • Coast Guard aircraft • International resources
- Helicopter use • Piloting a single helicopter • Water bombing aircraft • Piloting multiple helicopters and aircraft • SiTaC
- Environmental considerations • Cancellation of public waterway

In the 3rd edition, some minor editorial changes have also been made.

Andreas Sundberg and Anders Montan from the Coast Guard, Johan Wahlström from the Swedish Maritime Administration and Stefan Haggö, Leif Sandahl and Stefan Andersson from MSB have contributed material to the 3rd edition.



Fire hazard- forecasts

2. Fire risk forecasts

The web-based service Brandrisk skog och mark (www.smhi.se/brandrisk) is a decision support tool that can be used to assess fire risk and fire behavior. The service includes fire risk forecasts and historical fire risk data for the current calendar year, as well as weather data such as temperature, relative humidity, precipitation, wind speed and wind direction. The service is primarily aimed at municipal rescue services, county administrative boards and the forestry industry and can, for example, be used for preventive fire protection measures, planning firefighting efforts and issuing fire bans.

The fire risk values are calculated in squares of 2.8 km × 2.8 km and cover all of Sweden. There are 6-day forecasts, as well as hourly forecasts up to 48 hours ahead.

Below is a review and interpretation of the fire risk forecasts (FWI model) and meteorological data.

2.1 Fire risk values from the FWI model

In Sweden, the Canadian FWI model¹ has been used since 2000 for forest fire risk forecasts. Figure 1 provides a schematic illustration of the different sub-indices in the model. The model calculates the moisture content in three fuel layers (the FFMC, DMC and DC indices). The BUI moisture content (weighted average of DMC and DC) indicates the proportion of the fuel that is dry enough to be available for fire. BUI and ISI (the relative rate of spread of a fire) are then used to calculate the final fire risk value FWI index, which reflects the fire intensity.

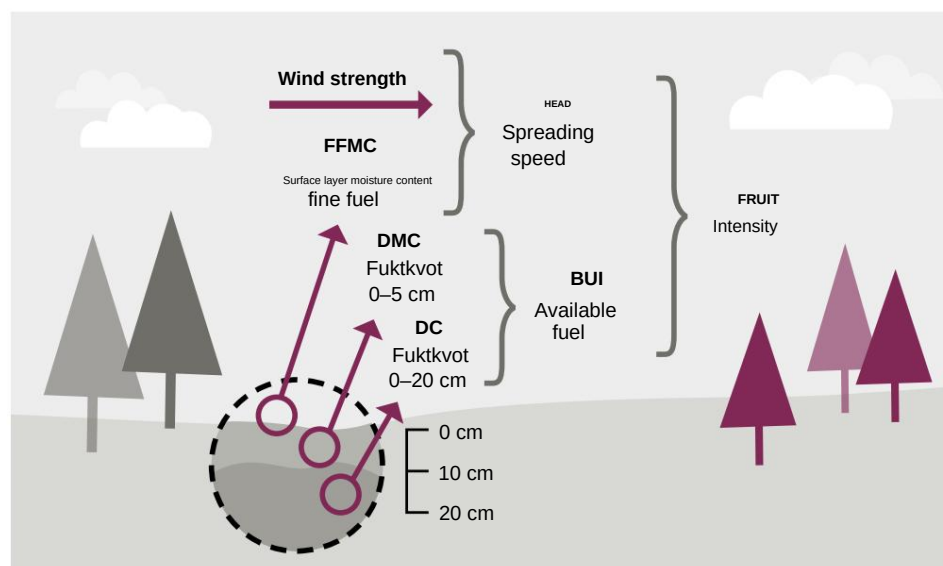


Figure 1. A graphical illustration of the various indices included in the FWI model.

1. Van Wagner CE. (1987). *Development and structure of the Canadian Forest Fire Weather Index System*. Forestry Technical report 35, Canadian Forestry Service, Ottawa.

Below are interpretations and rules of thumb for the FWI model's sub-indexes. Note, however, that the FWI model is developed based on Canadian vegetation, which may differ significantly from the fuel in Swedish forests. The interpretations and rules of thumb below should therefore be seen as a rough estimation method for fire behavior and spread rate.

2.1.1 FFMC (Fine Fuel Moisture Code)

FFMC represents the moisture in the top layer (about 1 cm) of finely divided fuel on the ground surface, for example leaves, needles and the upper part of the moss/fern.

Maximum water storage is less than 1 mm of water and this fuel layer therefore dries quickly.

When calculating FFMC, precipitation, relative humidity, wind speed, temperature and previous values for FFMC are taken into account.

The FFMC scale ranges from 0–101 (a low FFMC value indicates high humidity and a high value indicates dryness). An FFMC value above 75 usually indicates flammable conditions. A value above 90 is extremely flammable.

2.1.2 DMC (Duff Moisture Code)

DMC represents the moisture in a slightly deeper layer of the fuel on the ground (about 7 cm), for example the moss layer and the upper part of the humus layer. The water storage in this layer corresponds to up to about 15 mm of water.

When calculating DMC, precipitation, relative humidity, temperature and previous values for DMC are taken into account.

DMC usually varies between 0–150 for Swedish conditions, but has no upper limit. A low DMC value indicates high humidity and a high value indicates drought. If DMC exceeds 60, it indicates that the humus layer is dried out after a prolonged drought. Values above 120 are uncommon, but occur on average once a day per year.

High values of DMC mean an increased risk of intense and difficult-to-extinguish fires as well as an increased risk of lightning ignition during thunderstorms.

2.1.3 DC (Drought Code)

DC represents the moisture in thick compact organic soil layers (about 18 cm), such as the peat/humus layer. The water storage in the layer corresponds to about 100 mm of water.

When calculating DC, precipitation and temperature are taken into account, as well as the presence of current values for DC.

A low DC value indicates high humidity and a high value indicates drought (there is no upper limit, but the value is usually below 600 for Swedish conditions, except in the latter part of really dry summers).

High values of DC (over 350 for Canadian forests) mean an increased risk of difficult-to-extinguish smoldering fires in deep organic layers, with a risk of re-ignition.

High values of DC also mean an increased risk of lightning strikes during thunderstorms. Note that there may be some delay (up to several days) if the upper fuel layers are initially moist.

2.1.4 BUI (Buildup Index)

BUI indicates the proportion of the fuel that is dry enough to be available for combustion. The value is calculated as a weighted average of DMC and DC, with DMC being the most important.

The BUI scale generally varies between 0–150, but has no upper limit. A high BUI value indicates that a large proportion of the fuel will be used in combustion. Values above 150 are unusual for Swedish conditions.

2.1.5 ISI (Initial Spread Index)

The ISI represents a relative measure of the rate of spread (for Canadian pine forest) of a wind-driven fire. The index is calculated from the FFMC and wind speed.

The scale starts from zero; a low value corresponds to a low rate of spread and a high value corresponds to a high rate of spread (the ISI value corresponds approximately to the fire spread in meters per minute).

ISI values above 30 are quite unusual for Swedish conditions. If ISI exceeds 15, there is a risk of a top fire, i.e. the fire reaching the treetops. It also means an increased risk of aerial fires.

2.1.6 Fire Weather Index (FWI value)

The FWI value, calculated from the ISI and BUI, is the final index and indicates the fire intensity (expressed as power development per unit length of the flame front) of a fire that spreads with the wind. The FWI value reflects the expected fire behavior and the resources that may be required to extinguish the fire based on weather conditions and fuel drying.

Note that the model is developed based on Canadian pine forest, which can differ significantly from Swedish forest; different vegetation burns with different intensities.

The FWI value scale starts at 0 and has no upper limit.
In Sweden, the FWI value has been divided into 6 different risk classes, which are used, among other things, for communication to the public and the media, see Table 1 below.

Table 1. FWI index, FWI values and associated fire behavior.

Risk class (FWI index)	FWI value	Description
5E – Extremely high forest fire risk	28–	Requires prolonged drought, low relative humidity and some wind. The surface layer of the soil is extremely dry. The risk of ignition is very high and a fire will develop explosively. There is a high risk of a top fire, that is, the fire will reach the treetops. When it is 5E, private individuals should refrain from making fires or grilling in the forest and land.
5 – Very high forest fire risk	22–27	A fire will develop very quickly and violently. Top fires, i.e. the fire reaching the tree crowns, can occur. When it is 5, private individuals should refrain from making fires or grilling in the forest and land.
4 – High forest fire risk	17–21	There is a high risk that the fire will spread quickly in most types of forest land.
3 – Moderate forest fire risk	12–16	There may be a significant risk of fire spreading in certain types of forest land, for example in clearings.
2 – Low forest fire risk	6–11	In most types of forest, it is difficult for a fire to spread.
1 – Very low forest fire risk	<5	Little or no risk of fire spread in most types of forest land.

2.2 Meteorological variables with significant impact on fire risk

In Forest and Land Fire Risk there are also various weather variables (forecasts and historical data) that are of great importance for forest and land fire risk. Below is a brief overview of the most important variables.

- **Wind speed and wind direction** have a very large impact on fire behavior and spread rate. The main reason is that the wind bends the flames in the direction of the fuel. Increased wind can also increase the drying of the fuel and add more oxygen to the fire. Glowing and burning material can be carried by the wind and create flying fires. It is important to be aware that the wind can vary greatly on time scales down to minutes and seconds. The wind direction can also fluctuate greatly. In addition to the average wind, the gust wind (the highest wind speed during a 2-second period in the last hour) is important to keep an eye on in forecast systems. Frontal passages, especially cold fronts, can involve large changes in wind speed and direction.
- **Relative humidity** has a major impact on flammability and fire spread. For example, despite a prolonged drought, a high relative humidity can cause the surface fuel to become moist and thus make the fire spread moderate. The relative humidity is calculated based on the air temperature and dew point at a height of 2 meters, and indicates how much moisture content (water vapor) the air has in relation to the theoretical maximum possible (100% indicates completely saturated air, which can occur, for example, in rain or fog). Warm air can contain more moisture than cold air. If the relative humidity is less than 30%, critical fire situations with large fire spread and fire intensity can often be expected. In combination with high wind speeds, there can be a risk of aerial fires.
- **Temperature** affects the moisture content of a fuel and thus has an impact on fire risk. Temperature is also strongly linked to relative humidity and the fact that warm air can contain more moisture than cold air (the usually higher temperature in the afternoon means that the relative humidity is usually lower in the afternoon).
- **Precipitation** is a parameter that has a major impact on the drying out of the fuel and the risk of fire. Extra attention should be paid if there has been no precipitation for longer periods. In addition to forecasts for future precipitation, it may therefore also be relevant to analyze the accumulated precipitation for the previous days or weeks to investigate how much precipitation has fallen for different time intervals in the past. Note that a larger amount of precipitation over a shorter period of time (e.g. a heavy rain shower) does not lower the moisture content of more compact fuels as much as the corresponding amount of precipitation falling over a longer period of time (e.g. during sustained precipitation during a frontal passage).

- **Solar radiation** affects fire risk by drying out and lowering the moisture content of the fuel. Solar radiation depends on the time of year and day, as well as the current cloudiness. The amount of solar radiation an area is exposed to also depends to a large extent on the environment; for example, fine fuel often dries out faster in open environments than in forests, which is due to both higher solar radiation and a more wind-exposed location in open environments. Note that the sub-indexes in the FWI model do not take solar radiation into account, so it is recommended to keep track of this variable separately.

It is often a combination of several meteorological variables that contribute to a high fire risk. For example, warm air above 30 degrees with low relative humidity below 30% means a significantly increased risk from a forest fire extinguishing perspective. Warm air that is instead humid does not necessarily mean the same risk.

2.3 Interpretation examples for fire risk data

Figure 2 shows an example of fire risk data (daily values) from the Forest and Land Fire Risk service for the period Sunday, May 1 through Monday, May 2, 2022.

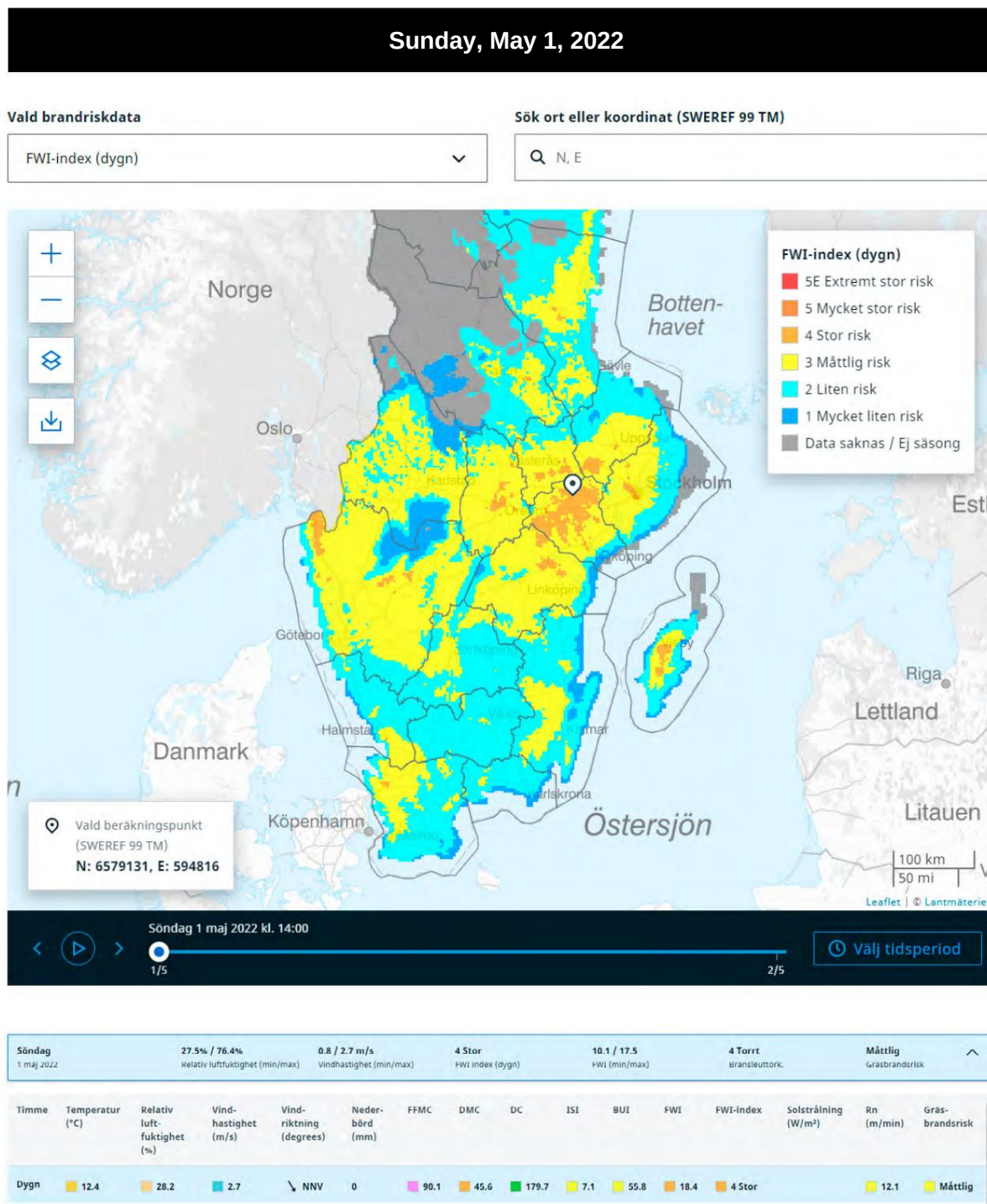
This example shows spring conditions when fire risk can be high early in the season, which is important to keep an eye on.

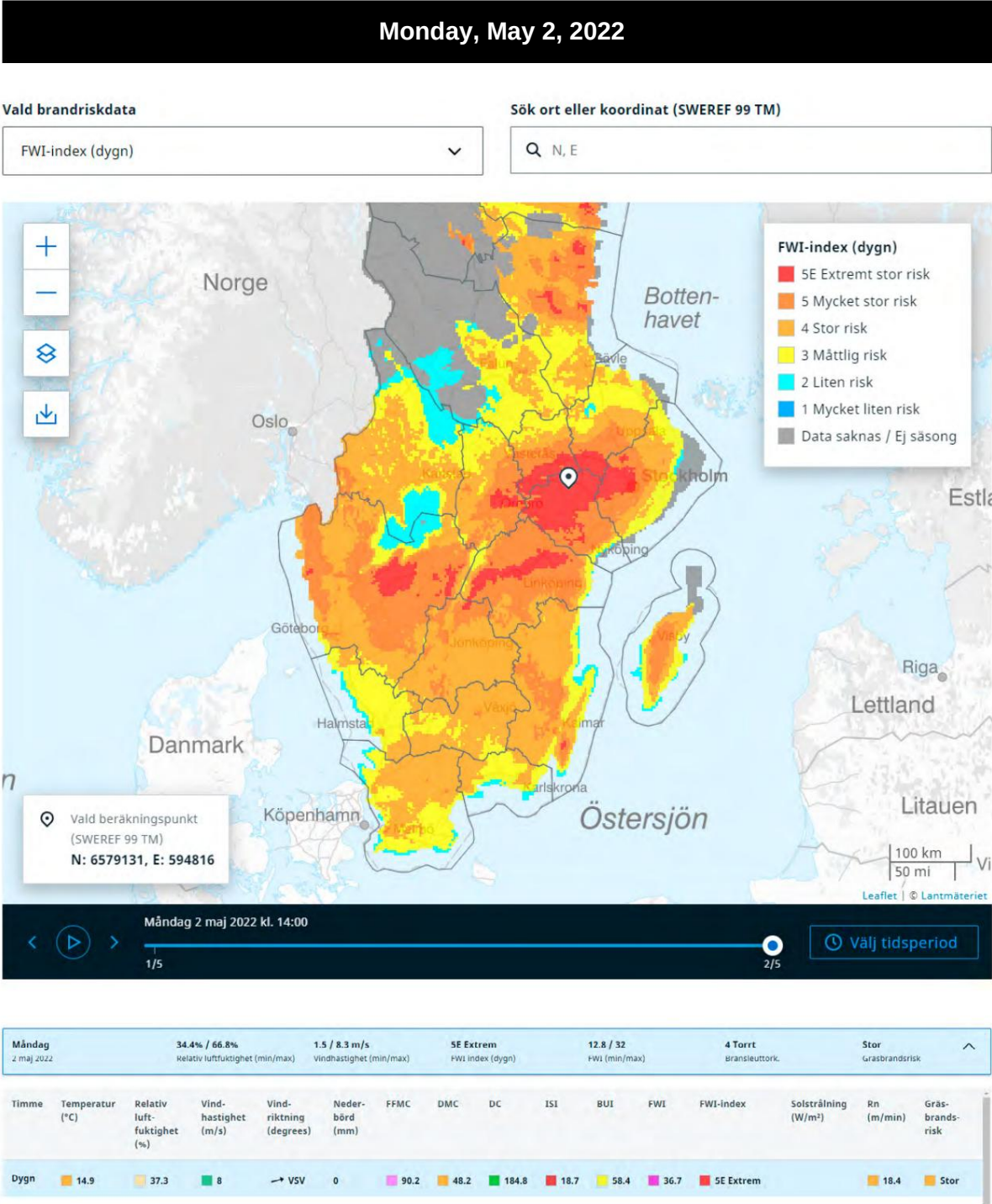
Both Sunday and Monday have similar characteristics in terms of drying out of the fine fuel. The FFMC on a daily basis is about 90 for the selected point, i.e. one can expect the fine fuel to be extremely flammable on both Sunday and Monday.

The air is fairly dry for both days with a relative humidity of about 28% for Sunday and about 37% for Monday. What distinguishes the days is that the wind speed is considerably higher on Monday (an average wind according to the daily model of about 8 m/s compared to just over 2 m/s for Sunday). The wind during Monday means that the spread rate ISI amounts to just over 18, which means a risk of peak fires and aerial fires. Since the ISI is so high during Monday (due to high wind speed in combination with fairly low relative humidity), the FWI value will also be very high (just over 36), i.e. an FWI index of 5E (extremely high fire risk). Violent fire events with high spread rates in the fresh and strong gusts of wind are to be expected and the fires could be resource-intensive.

It is noteworthy that DMC and DC do not have such high values (45 and 180 respectively), This is reflected in the fact that the deeper fuel layers have not yet dried out as much during the spring. Thus, somewhat more superficial fires are to be expected.

During Monday, it should also be taken into account that a fire could spread from grass to forest as the risk of grass fires, under the influence of the high wind speed, is high with a spread speed (Rn) of approximately 18 meters per minute.





Detection of forest fires

3. Detection of forest fires

3.1 General

In addition to human suffering, forest and land fires also cause costs for society and environmental effects – both positive and negative. One of the most important tools for mitigating the negative effects of forest fires is early detection of the fires.

The concept of early detection of the fire, combined with rapid first aid, intervention while the fire is small has long been well proven when it comes to fighting forest fires. If forest fire extinguishing is started at a later stage, there is a risk that the fire has grown in size, developed a larger fire front and spread to vegetation with a worsening fire course. The extinguishing work then risks becoming very resource-intensive and that the rescue service risks ending up behind schedule in relation to the fire. Thus, forest fire detection is a very important component in the field of forest fire extinguishing.

3.2 Forest fire detection in Sweden

Already at the end of the 19th century, organized forest fire detection began in Sweden, in the form of an expansion of forest fire towers. During the 1930s, the first trials of forest fire flights were carried out in Jämtland based on experiences from earlier trials in Finland and Canada. During this period and during the Second World War, the network of forest fire towers was also expanded on a large scale, to a nationwide system. After the Second World War, the fire function for forest fire towers south of the Karlstad–Gävle line was discontinued. The reason was that the number of fires detected in southern Sweden was very low. During the 1950s, trials and then implementation of forest fire flights began due to the increasingly increased costs for forest fire towers. In line with the expansion of forest fire flights, forest fire towers were discontinued to a corresponding extent. During the 1960s, the last forest fire towers in Sweden were discontinued².

Currently, the following two detection methods are mainly found in Sweden:

- Forest fire surveillance with forest fire aircraft.
- General public.

2. Larsson L. (2002), *Forest fire surveillance by air*, PM 2002-10-22, Unit for damage control activities, SRV, Karlstad.

Forest fire surveillance by forest fire aircraft is carried out along a number of predefined routes distributed around the country. The overall goal of the forest fire surveillance by aircraft function in Sweden is to detect and determine the position of forest fires as early as possible in order to thereby minimize the spread of the fire and thereby costly rescue efforts and major property losses. MSB is responsible for overall direction, financing, follow-up and final evaluation, and final reporting of the results of the operation. The County Administrative Board, which has production responsibility for forest fire surveillance by aircraft, decides within the framework of MSB's guidelines, primarily whether surveillance by aircraft shall be conducted in the county. The County Administrative Board then decides on the scope of the surveillance, for example with regard to the route of the flight routes and how often surveillance shall be carried out. The County Administrative Board also draws up a special plan for the surveillance, for monitoring the results in its county and for annual reporting.

The term public generally includes all non-organized detection activities and can, for example, include tourists in forest areas, berry pickers, hunters, commercial aircraft, etc. Statistics from both Sweden³ and Canada⁴ show that the general public detects a majority of forest fires, even in areas with extensive forest fire surveillance aircraft. The proportion of fires detected by forest fire surveillance aircraft is inversely proportional to population density³, but with mobile telephony, the general public has most likely come to play an increasingly important role in the early detection of forest fires, even in sparsely populated areas.

3.3 Additional detection means

Below are two examples of additional methods and are not a complete list of existing detection methods. For a more comprehensive account of different detection methods, a feasibility study on future detection of vegetation and forest fires is recommended⁵.

3.3.1 Satellite detection

With the development of operational satellites for observing the Earth's surface, tools were provided for continuous scanning of the Earth's surface. Depending on the orbit, observation technology and platform technology, information about the Earth's surface can be provided within a very short period of time for most locations on Earth. Sensors on satellites can be used, among other things, for observations of the Earth's vegetation - where data regarding the moisture content of the vegetation can be included in fire risk models - but also for the detection of forest fires. Detection of forest fires using satellites has been occurring in a number of different countries such as the USA, Australia and Finland for several years. In Sweden, satellite detection of forest fires has been tested for several years, but was put into operation in 2022. The disadvantages of satellite detection are primarily that the fire must be of a certain size or intensity to enable detection and

3. Hertzberg Å & Lundqvist M (2022). *Forest fire surveillance flights – a description and evaluation of the operation*. Report 1898, MSB, Stockholm.

4. Kourtz P (1987). *The need for improved forest fire detection*, Forestry Chronicle, augusti 1987, Kanada.

5. Björck J (2021). *Pre-study – future detection of vegetation and forest fires*. Report 1725, MSB, Karlstad.

that certain weather conditions limit its use (especially cloudy weather). In the Swedish satellite trials, it has been shown that the smallest fires detected were of the order of a controlled burn on a farm or a residential fire⁶. In 2020 and 2021, tests and evaluation of real-time satellite detection of fires have been carried out in Sweden by MSB in collaboration with SMHI and 18 municipal rescue regions/rescue services. In the event of satellite detections, the participants in the test have received notifications in the form of e-mail and text message.

The satellite detection was made possible by two polar weather satellites (Suomi-NPP and NOAA-20) equipped with so-called VIIRS instruments, where both infrared and visible light are used to detect active fires. Evaluation⁷ of the test showed that just over 60 percent of the satellite detections could be linked to fire rescue efforts, of which just over half were vegetation fires. Furthermore, 25 percent of the satellite detections that could be linked to vegetation fires occurred before the alarm was received by SOS Alarm. The majority of these fires were in sparsely populated municipalities in Norrland, which underlines the technology's potential for early detection in primarily sparsely populated areas. A few satellite detections were not fires (e.g. sun glare from roofs) and for a relatively large proportion, about 30 percent, the cause of the detection was unknown (a large proportion of these are assumed to be controlled fires, such as nature conservation burns or grass/rice fires in the spring). Feedback from the rescue services showed that the positioning of the satellite detection is of relatively high accuracy (maximum 600 meter error, usually significantly less than that) and that the technology is a good complement to other detection techniques. A limitation, however, is that detections can only see when the satellites pass over Sweden (for these VIIRS satellites it is at night/early morning and early afternoon respectively) and that cloudiness is a limiting factor. Further information about the satellite detection and evaluation, including analysis of the impact of clouds on satellite detection, is published in a thesis at Uppsala University⁸.

In 2022, satellite detection will be put into operation and will be available from May in the Forest and Land Fire Risk service (www.smhi.se/brandrisk).

3.3.2 Stationary cameras

During the 1990s, development of stationary camera systems for the detection of forest fires began. The systems are well suited for areas where early detection is desirable. The system has – in addition to the possibility of early detection – advantages such as continuous monitoring of the area and is therefore not dependent on overflights or passages, detection around the clock and the system is not as weather-dependent as satellite systems as long as the camera and the target are below cloud base. The camera systems exploit, among other things, the fact that fires that occur in ground vegetation or in moist fuel give rise to a fire plume of a white character.

6. Björck J (2021). *Pre-study – future detection of vegetation and forest fires*. Report 1725, MSB, Karlstad.

7. Letalick M & Andersson S (2022). *Follow-up of satellite detections of forest and vegetation fires. The period January through October 2021*. MSB1939 – April 2022.

8. Letalick M (2022). *The impact of clouds on satellite detection of vegetation fires in Sweden*. Degree thesis at the Department of Geosciences at Uppsala University, ISSN 1650-6553; 539 urn.kb.se/resolve?urn=urn:nbn:se:uu:diva-472777.

The disadvantages of a camera system are that it is stationary and only covers a certain, limited geographical area, and that the costs risk becoming considerable if larger areas are to be covered. The latter underlines the fact that the system is particularly suitable in areas where early detection is desirable, due to, for example, sensitive objects or high fire potential. Camera detection occurs in a number of different countries, such as Germany, Estonia, Lithuania, Spain and Portugal.

During the summer of 2013, MSB tested camera detection of forest fires from a telecommunications mast in Sunne municipality. The system tested included optical sensors in the form of a camera. The camera could rotate 360° and make a sweep in 6 minutes during the day and 12 minutes at night. The camera could cover an area of up to 700 km². The detection itself was carried out by analyzing images from the sensors to detect fire smoke during the day or fire smoke illuminated by near infrared radiation from the fire during the night. In the event of a fire, the system sent an alarm to an operator who in turn had the task of determining whether the incoming alarm was a probable fire or a false alarm in the form of, for example, a dust cloud. The tests showed that the system detects fires at an early stage and the associated position indication has a high accuracy. The disadvantage of the system was primarily the limited geographical coverage, which has consequences given the large areas to be covered in Sweden.



The forest fire and its course

4. The forest fire and its course

4.1 The parts of a forest fire

The concepts for the different parts of a forest fire must be known to everyone involved in a forest fire. They are needed as a reference when directing personnel, helicopters, etc.

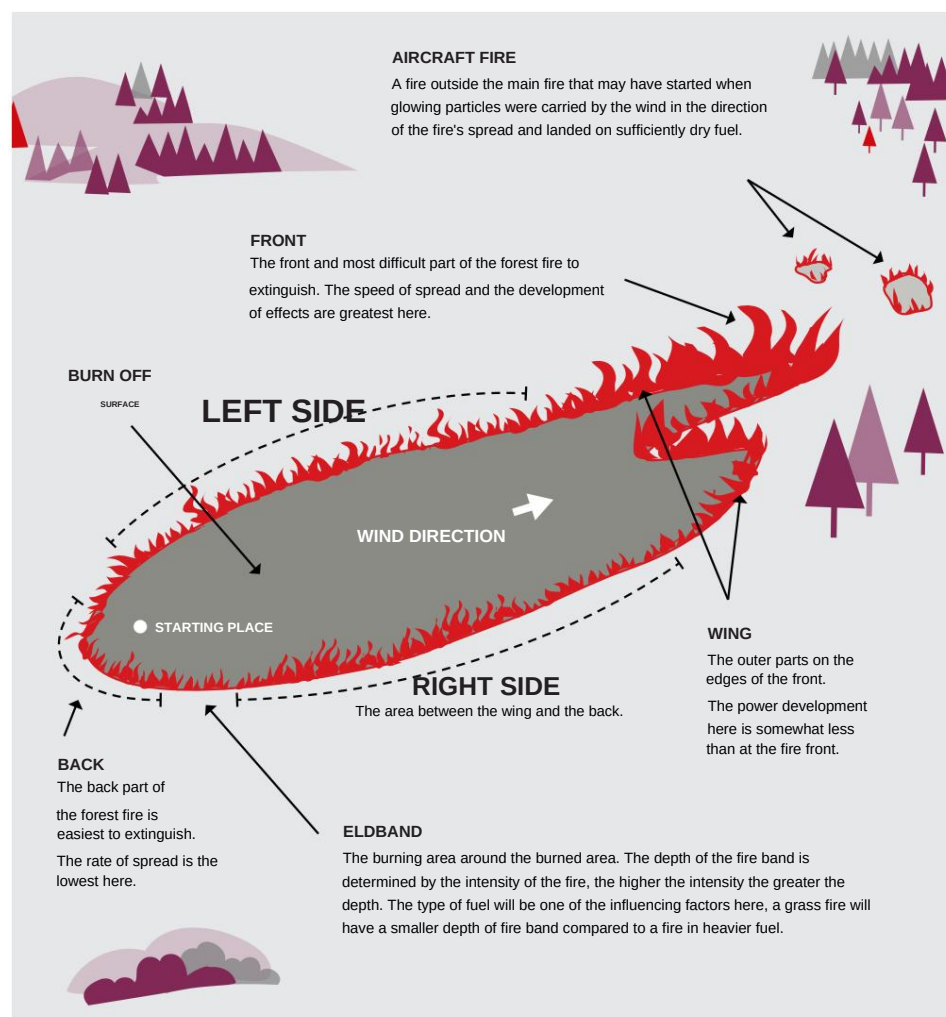


Figure 3. The parts of a forest fire.

The appearance of the fire area will largely depend on the prevailing wind strength and direction. In a forest fire in uniform vegetation and topography and with increasing wind, the fire area will increasingly take on the shape of an ellipse.

4.2 Different types of forest fires

It is important to know the four main types of forest fires as well as their behavior and spread rates in order to be able to fight them as effectively as possible.

The values for spread rates given below in the section are examples; there are forest fires that burn faster or slower. The division into categories below is based on the different fuel layers, but when it comes to spread rates, weather and terrain factors will also come into play. In most cases, the wind will have a strong impacting effect.

4.2.1 Peat fire (fire in deeper soil layers)

Peat fire is a smoldering fire that can occur in an oxygen-poor environment below ground level after a low or high wildfire has burned through. The fire spreads via roots and dead vegetation.

A fire can go very deep, depending on how thick the layer of dead vegetation is. There is a risk that the peat fire will turn into a running fire again. A peat fire can continue for a very long period and the spread is slow, often only a few centimeters per hour. Peat fires are only possible in forest lands that have peat soil or under a thick layer of peat in peatlands, for example in an older spruce forest.

4.2.2 Low running fire

Low-speed fires are the most common type of forest fire. The majority of all forest fires start as low-speed fires, which means that they burn in ground vegetation, dead vegetation and low bushes. An example of a spread rate range is 0–10 meters/minute. In areas where grass dominates, the spread rate can be up to 15–30 meters/minute.

4.2.3 High running fire

A high-spreading fire is a fire in the ground vegetation, in lower branches and in individual tree tops. What distinguishes a high-spreading fire from a top fire is that in a high-spreading fire only individual tree tops are involved and it is strongly dependent on the fire in the ground vegetation. An example of an interval for spread speed is 10–20 meters/minute.

4.2.4 Top fire

A top fire spreads among the higher fuel layers, from treetop to treetop. The prerequisites are that there is a strong wind, that the distance between the treetops is not too far, that there is continuous fuel from the ground up towards the treetops and that the amount of available dry fuel in the ground and field layer (the layer above the ground layer) is sufficient. As a rule, in addition to a fire with sufficiently high fire intensity and depth of the fire band, a low moisture content of the tree crown in combination with a short distance up to the lowest branches is also required to achieve a top fire. An example of an interval for spread speed is 20–50 meters/minute. A fire must go from a low running fire to a high running fire before it transitions to a top fire.

Peak fires are the type of fire that we associate with extreme forest fire events and they are found in both wind-borne peak fires and fire plume-dominated fire events. The term crown fire also occurs as a designation for this type of fire.

NOTE: When a fire spreads to a higher fuel layer, for example when a low-level fire transitions to a high-level fire, the fire will undergo an acceleration phase with increasing spread rate and power development.

4.3 Extreme forest fire events

Extreme forest fire events typically involve fire in the higher fuel layers and can be divided into two different types of fire events:

- wind-borne peak fires
- fire plume-dominated fire progression.

In these two types of fire events, phenomena such as fire whirls and flying fires will also occur.

4.3.1 Wind cage top fire

The wind-borne top fire requires above all strong winds and continuity of fuel both vertically and horizontally in order to occur. In case of weaker winds or interruptions in fuel continuity, the horizontal spread of fire in the higher fuel layers will stop and only a few trees will be affected. A wind-borne top fire is a forest fire where flame radiation and convection ignite the higher fuel layers with the help of the strong wind.

Wind profile

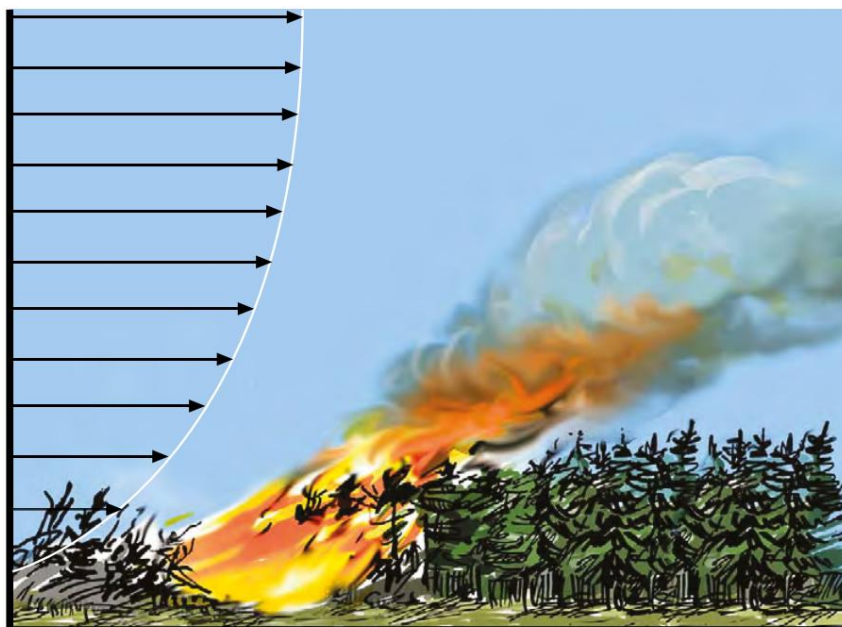


Figure 4. Wind-borne treetop fires can occur when wind speed increases and fire gases from burning trees ignite adjacent treetops. Illustration: Per Hardestam.

The fire has a clear direction of spread and a fire area of a tapered, elliptical character. In this type of forest fire, fire whirls and flying fires often occur; however, the throw distance of the flying fires is not as long as in the fire plume-dominated variant.

There are usually two forms of wind-borne top fires. The first form is top fires where the spread of fire in the higher fuel layers is dependent on the fire in the lower fuel layers, and if the fire in the lower fuel layers stops or subsides, the fire in the higher fuel layers will also stop.

In the second form, the power development from the fire in the higher fuel layers is so high that it is completely independent of the fire in the lower fuel layers.

4.3.2 Fire plume-dominated fire progression

Among the most extreme fire processes that can occur in a forest fire is a fire plume-dominated fire process, i.e. a fire process where the fire plume itself has the strongest impact. For a fire plume-dominated fire process to occur, a high fire load, very low fuel moisture content and unstable air masses are required. This type of fire process is associated with weaker winds than the corresponding wind-borne top fire – usually less than 8 meters/second – and a clear fire plume, also called a convection plume.

Wind speeds are often higher at ground level than higher up.

Wind profile

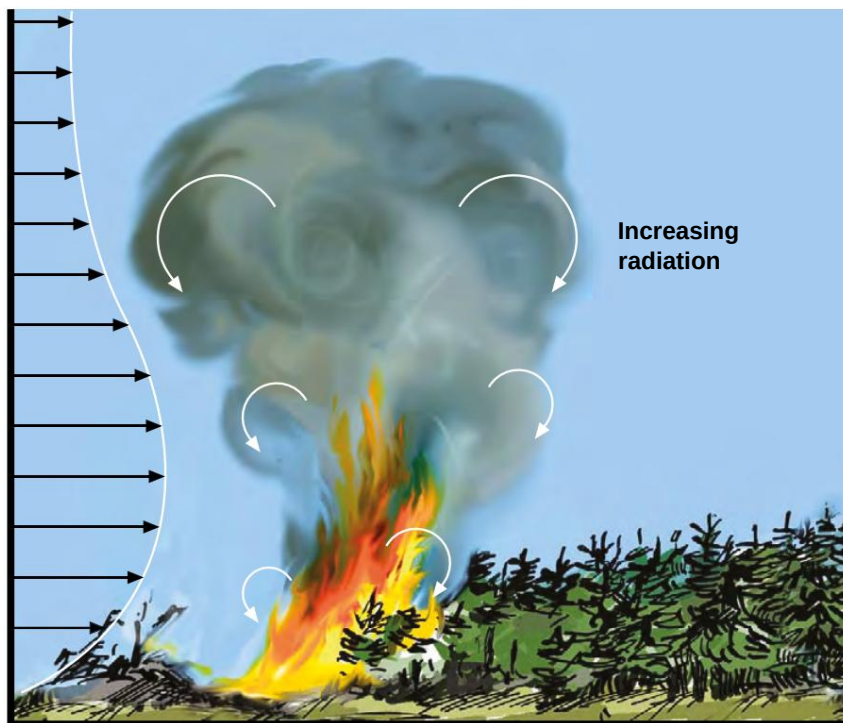


Figure 5. For a fire plume-dominated fire process to occur, a large amount of fuel and very low fuel moisture content are required. Illustration: Per Hardestam.

The fire plume-dominated fire process spreads in two different ways: firstly through convection and flame radiation from the fire plume, and secondly with the help of strong downward winds, which cause fire spread in different directions. The resulting wind speeds can be very high and can topple or break trees.

The plume-dominated fire progression burns with a higher power development but has a lower spread rate compared to the wind-borne top fire. However, the violent combustions that can occur in the plume-dominated fire progression still give rise to higher spread rates than in the wind-borne top fire. The spread of fire in a plume-dominated fire progression is often difficult to predict, since sudden and violent fire spreads in widely different directions occur.

The fire plume-dominated fire progression continues for a maximum of a number of hours and then ceases. This type of fire progression often starts in the early afternoon when the relative humidity is lowest, the temperature is highest, the availability of dry fuel is greatest and the probability of unstable air masses is greatest.

The high fire plume means increased throw distances for aerial fires as the plume (and burning or glowing material) reaches higher into the atmosphere.

Any attempts to extinguish the fire during the plume-dominated fire process will most likely be futile and too risky. The best thing to do is to let the ground personnel retreat and wait for the fire process to stop and for more favorable weather conditions to arise. However, airborne resources with greater water capacity can possibly be deployed to protect buildings, for example.

Emergency services personnel on site receive an early warning based on the appearance and development of the fire plume, and color changes, strong growth and turbulent movements are clear warning signs. The fire plume can be advantageously monitored from the air, where any changes can be seen early and clearly. Any fire vortices in the area can indicate unstable air masses and very low humidity levels can also serve as a warning signal and be taken into account in a risk assessment.

4.3.3 Fire whirls

Fire whirls generally require unstable air masses and high fire load to occur. Fire whirls can, for example, occur in connection with a cold front passing over the fire area or on the leeward side of a hill. The fire whirls themselves cause a strong suction of air at ground level, tornado-like winds, increased effect development on the fire, increased flame length, increased spread speed and aerial fires. The greatest risk with fire whirls is the aerial fires that they spread in the terrain at a shorter distance from the fire front. Fire whirls are usually less than one meter in diameter and then do not pose any major risks, but with increasing size the risks described above increase.

4.3.4 Aircraft fires

A flying fire is a forest fire that has arisen outside the main fire area and that may have arisen by burning or glowing material being carried by the wind in the direction of the fire's spread and subsequently igniting fuel on the ground surface or in higher fuel layers. The glowing material may be in the form of a leaf, a twig, a cone or a piece of bark. The actual ignition of fuel on the ground surface or in higher fuel layers may be delayed in time as the fuel may need to be dried out and heated up by the burning or glowing material. Some time delay can therefore be expected before the actual flying fire occurs. The occurrence of flying fires is usually a clear indicator of extreme forest fire events.

Since aerial fires can greatly worsen the course of a fire and make extinguishing work more difficult, fighting them should be prioritized. Flying resources can be used to advantage in fighting aerial fires.

The throw distance of aerial fires can vary greatly from a few meters to several kilometers. Aerial fires can occur as a result of fires in the lower fuel layers but are more common as a result of fires in the higher fuel layers and fires in piles of logging residues.

4.3.5 Combustions

The phenomenon of forest fires burning together (coalescence) is an example of a very extreme fire process, perhaps even the most extreme one that can be experienced in a forest fire. Separate forest fires influence and attract each other, both protective burns and counterfires as well as the main fire itself. The larger and more intense the fire, the greater the impact it has on more distant fires.

The phenomenon can occur during a single forest fire where separate fire fronts are positioned against each other – or in forest fires and burning that have resulted in aerial fires at a shorter distance from the fire front and which have subsequently burned together with the forest fire with violent force.



Assessing the spread and extent of a forest fire

5. Assessing the spread and extent of a forest fire

Below are a number of simple methods or rules of thumb for how the spread or progress of a forest fire can be roughly estimated and assessed. Due to the simplicity of the methods and the fact that they do not require any major requirements for habit or experience, they are well suited for fire officers, regardless of their level of knowledge, and for use on site in an initial stage of the fire.

Perhaps the simplest method for estimating the spread and extent of a forest fire is to mark the location of the fire front on a map at two known times and measure the distance between the two positions to obtain a rough estimate of the spread rate. However, this assumes that all factors are constant: wind speed, wind direction, topography, etc. It is therefore a rough estimation method, but it is very field-based and useful.

5.1 The influence of wind on the speed of fire spread

In southern Europe in particular, a rule of thumb is used where the fire front is assumed to have a spread speed of 3% of the wind speed. However, this rule of thumb does not take into account changes in vegetation or the slope of the terrain.

Table 2. Examples of fire front spread speed and propulsion at different wind speeds.

Speed				The progress of the fire front		
Find		Fire front		Meters at the following time intervals		
Kilometer/ hour	Meter/ second	Meter/ hour	Meter/ minutes	15 minutes	30 minutes	60 minutes
10	3	300	5	75	150	300
20	6	600	10	150	300	600
30	8	900	15	225	450	900
40	11	1200	20	300	600	1200
50	14	1500	25	375	750	1500
60	17	1800	30	450	900	1800
70	19	2100	35	525	1050	2100
80	22	2400	40	600	1200	2400
90	25	2700	45	675	1350	2700

5.2 The appearance of the fire area is affected by the wind

Wind will strongly affect the appearance of the fire area. At wind speeds below 0.3 meters/second, the fire area will have a circular shape with uniform spread in all directions, provided that the topography and vegetation are similar in the area.

At higher wind speeds, the fire area begins to take on a more elliptical spread. The higher the wind speed, the more extended the fire area becomes. The fire area spreads to different degrees in different directions depending on the wind strength.

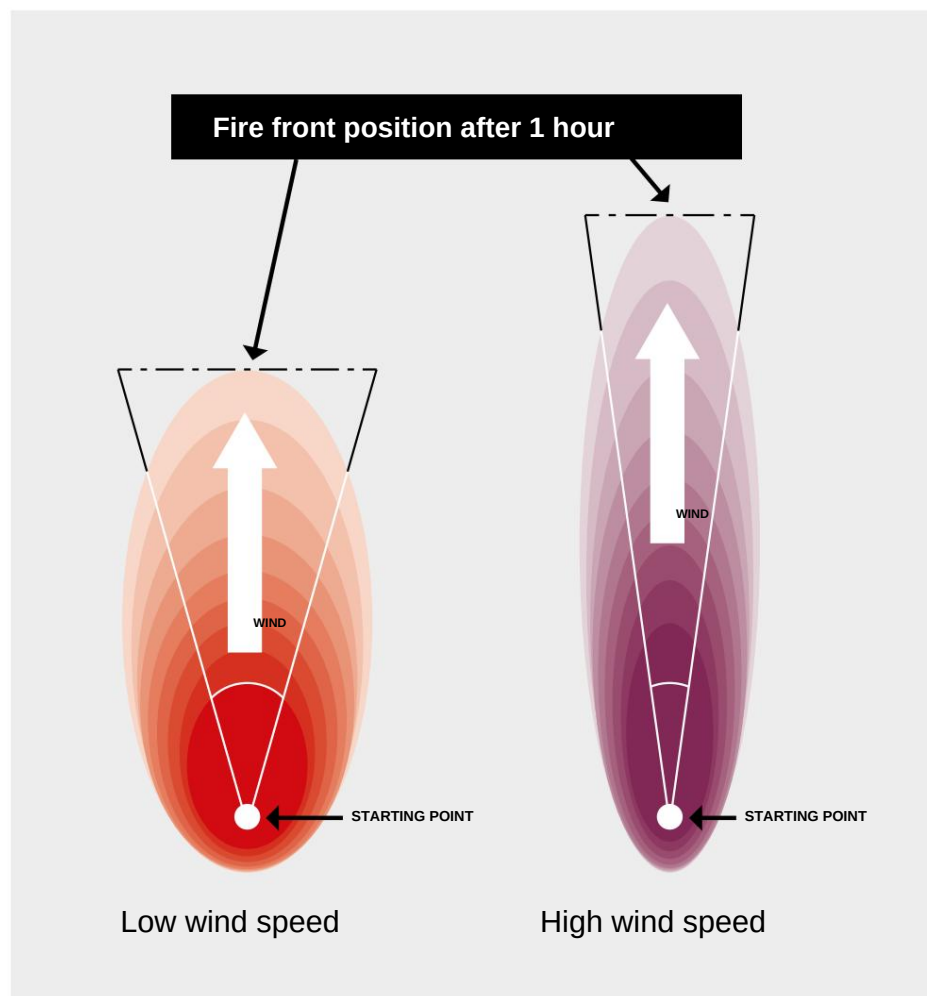


Figure 6. Forest fire spread at different wind speeds.

At lower wind speeds, the fire will spread not only in the wind direction but also against the wind direction at the back of the fire and to an even greater extent out from the flanks and wings. With increased wind speed, the greatest spread will occur at the fire front. In order to assess the spread of the fire in the wind direction, the angle of the spread cone is of great interest.

As a suggestion, you as a fire officer can produce a transparent template with different dispersion cones which is then placed on a map with an appropriate scale.

Table 3. Angle of the dispersion cone at different wind speeds.

Wind speed (meters/second)	Cone angle (degrees)
1	70
2	50
3	40
4	30
5	25
6	20
7	20
8–10	15
11–20	10

5.3 Fire history

The fire history of an area can be important to know. It is advisable to seek advice from someone who has been involved in a forest fire in the specific area before. The fire history can, for example, describe how large the forest fires have been, what the flame heights have been and whether there have been any peak fires. The fire history can be good to know because the course of a fire in an area tends to be similar under similar weather conditions.

It is not necessary to know the exact weather conditions at a previous fire, FWI values (Fire Weather Index) etc. but only in broad terms. If we have a summer with similar conditions to a previous summer, it may be good to remember the fire behavior in the specific area, for example.

The advantage of fire history is that it links the weather aspect with the fuel aspect, provided that the vegetation type and amount of vegetation have not changed to any great extent in the area. As forest fires become more frequent, this tool will also become increasingly important.

5.4 FBP (Fire Behavior Prediction)

Using the Canadian model FBP (Fire Behavior Prediction), spread rates can also be determined. Input values include a series of index values from FWI (www.smhi.se/brandrisk) but also parameters such as fuel type, wind speed, wind direction, slope, direction of the slope, height above sea level, etc. Values that can then be read out include spread rate and burned area (the appearance of the fire area).

However, using the model requires a certain level of skill and experience, for example in assessing which fuel type should be taken into account. The model is therefore not developed further here, but we instead refer to existing literature: See, among others, KG Hirsch, *Canadian Forest Fire Behavior Prediction (FBP) System: user's guide*, Canadian Forest Service.



Forest fire fighting

6. Forest fire fighting

The main firefighting takes place on the ground and it is here that the fire is finally declared extinguished.

In the case of smaller forest fires or forest fires with low fire intensity, the fire is fought. The fire is usually fought with a so-called direct attack, where the rescue service personnel go directly to the fire band. In the case of forest fires with high fire intensity or spread rate, the containment lines are carried out at a distance from the fire band through a so-called indirect attack. In the case of larger forest fires, the rescue service may use both direct and indirect attacks. In the more low-intensity sections of a forest fire, direct attacks are used and along the high-intensity sections, indirect attacks are used.

6.1 Direct attack

Direct attack means that emergency personnel go directly into the line of fire. The method is used for smaller forest fires and forest fires with low fire intensity or low spread rate.

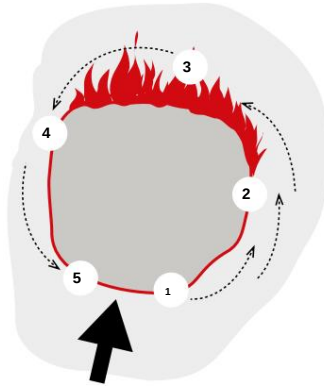
The advantages of this method are that the fire is usually prevented from spreading to the higher fuel layers, that protective burning (which can pose safety risks) does not need to be carried out and that the size of the burned area is minimized.

The disadvantages are that personnel are exposed to smoke and heat, that any fire barriers cannot always be used because it is the position of the fire band that determines the placement of the boundary line, and that a flank or wing can suddenly become a fire front in the event of a wind shear. Also consider the need to secure the outer edges of the fire to avoid re-ignition.

Whenever possible, direct attacks are carried out first, as this has the greatest chance of success and in the event of any flare-ups, rescue personnel can quickly take refuge in the burned, black area that is directly adjacent. In this case, strive to seek open areas to minimize the risk of falling, fire-damaged trees.

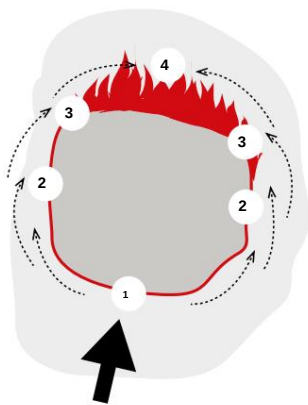
In connection with the direct attack, an immediate measure may also be to carry out a firefighting operation against the most intense section along the fire belt, which is fought separately to keep the fire down and gain time to create containment lines.

The attack itself can be carried out in three different ways:



1. The emergency services personnel begin the attack at the back of the fire (1), continue along the most critical flank (2) and the fire front (3), move on to the other flank (4) and finish at the back (5).

Figure 7. Direct attack with a fire engine.



2. The emergency services personnel begin the attack at the back of the fire (1), continue on both sides of the fire along the flanks (2) and wings (3) and then finish at the fire front (4). If necessary, emergency service personnel can – if the situation is deemed safe – begin the attack at the fire front immediately.

Figure 8. Direct attack when you have several fire engines at hand.



3. The third method is a mobile attack. The mobile attack can, for example, be used for smaller grass fires or in terrain that allows off-road driving. The method involves pulling out a shorter length of narrow hose and a jet pipe from an off-road fire engine. A firefighter mans the jet pipe and the fire engine begins to roll along the fire line. The firefighter then extinguishes the fire line while driving.

Figure 9. Mobile attack. Illustration: Per Hardestam.

The method can be further refined by driving with several fire engines that start on each side and meet halfway. In addition, the rescue service personnel do not have to drive along the fire belt but can drive through the burned area and attack the fire front from behind. This saves time. The mobile attack can also be used to suppress a forest fire that is approaching a road/fire road, which increases the efficiency of the fire road.

In some situations, it is not possible to begin the direct attack in the rear due to the nature of the terrain or threatened values in the direction of spread of the fire front. The direct attack may then begin at a more critical section of the fire and it is important to ensure that the fire does not bypass the containment line.

6.2 Indirect attack

Indirect attack means that the boundary lines are created a longer distance from the fire. The method is used for larger forest fires with high power development and rapid spread rates.

The advantages are that personnel are not exposed to heat and large amounts of smoke, and that natural fire barriers (lakes, roads, etc.) can be utilized when constructing the boundary lines.

The disadvantages are that the attack involves longer boundary lines and increased time consumption. Long boundary lines mean an increased number of places where the fire risks getting past the boundary line and that the increased time consumption means a risk that less favorable weather conditions and increased fire intensity may occur. In addition, a larger amount of fuel will be found at the boundary lines that has not yet burned and which, with worsening fire conditions, will pose a risk. Another disadvantage is that larger areas will be burned. However, it should be remembered that in the case of larger, violent and rapidly spreading forest fires, the indirect attack is often the only realistic alternative.

The indirect attack can be advantageously supplemented with protective burns. Once firefighters have established the containment lines. Protective burns aim to widen the containment lines, ensure that they remain contiguous with the fire itself, and to get rid of potential sources of risk in the form of vegetation that has not yet burned out.

In many large and intense forest fires, the fire service uses both direct and indirect attacks. Direct attacks are used in the rear and along the flanks, and indirect attacks are used along the wings and fire front. Furthermore, the fire service may have to use indirect attacks on an active forest fire during the day, but switch to direct attacks at night when the fire is less active.

There is also a middle ground between direct and indirect attack. In this attack, the boundary line is not constructed directly at the fire line as in the direct attack, but at a distance of a few meters to 15–20 meters from the fire line. In this way, most of the advantages of the direct attack can be utilized without being exposed to the worst of the heat or smoke. The fire can be allowed to burn along with the boundary line or protective burning is carried out. Protective burning is then carried out at the same time as the construction of the boundary line, unlike in the indirect attack where protective burning is usually carried out after the boundary lines have been completed. It may facilitate subsequent work if the fire is allowed to burn along the containment line.

The method can be used advantageously along flanks and wings in high-intensity forest fires where direct attack is not possible. In these cases, better weather conditions are often awaited before attacking the fire front. Also consider the risk that a flank or wing could suddenly become a fire front during a wind shear.



Boundary lines

7. Boundary lines

Limitation lines can be implemented in different ways, depending on water availability, among other things. Usually, limitation lines are implemented in the form of hose systems, fire streets and in combination with protective burning. The methods are not mutually exclusive; good results can be achieved if you combine them.

The boundary lines should be as straight as possible, so that it is easier to see them and check that they are holding. In the case of a forest fire on a slope, the boundary line is placed on the opposite side of the crest. This means that the wind direction is on your side and the speed of fire spread in downhill terrain (negative topography) is low.

7.1 Hose system or sprinkler system

In the direct attack, water is advantageously used because of its cooling effect. effect on the flames. When it comes to using water in the indirect attack, the benefits of water are not as great as the evaporation of the water. In addition, as air temperature increases, the application of water along terrain sections will have a decreasing effect.

NOTE: Not all forest fires need to be completely surrounded by a hose system. For example, fire lanes alone can constitute a boundary line in certain terrain sections.

The hose deployment will be progressive, with outlets in the form of branch pipes for narrow hose and jet pipes every 50 meters. The hose deployment starts at an anchor point, for example at a road or in the burned area at the back of the fire, and can either go along one flank or along both flanks. An anchor point is usually a low-lying point in the terrain behind the back of the fire. The anchor point aims to anchor the boundary line to prevent the fire from creeping past the boundary line.

Another option is to use a mobile sprinkler system, which can be laid out in the forest terrain to create a wet boundary line.

When placing the hose system or sprinkler system, existing and natural fire barriers in the form of roads and paths should be utilized as far as possible.

This will make the work easier and faster. The boundary lines can also, if necessary, be drawn through more finely divided fuel such as grass and bushes.

Avoid drafts through fuel of a compact nature to reduce the risk of the fire jumping the containment line.

A boundary line consisting of a hose system or sprinkler system in the terrain should not be considered a complete containment line unless it passes through very light fuel or the fire has burned to the containment line and then been extinguished. The containment line should generally be supplemented with a fire escape or a protective burn should be carried out adjacent to the containment line.

Please note that there may be quite large pressure differences in the hose system due to differences in altitude in the terrain. In cases where the pump is located at a lower point, a pressure booster pump may be needed. In cases where the pump is located at a higher altitude, the opposite problem may arise, i.e. too high pressure in the hose system.

7.1.1 Water use

Spreading jet is applied in sweeping movements to the base of the fire and to the fuel right next to the fire band, not directly to the flames. Closed jet is used for forest fires that have reached high up in the vegetation, to tree branches, etc.

The water jet operator should be careful not to spread the embers from a burned-out area to an area not yet affected by fire. It is best if the water jet operator stands on a surface that is not affected by fire and sprays towards the affected surface. The rescue service personnel can be provided with tools such as axes, shovels, etc. to assist the water jet operator in extinguishing the fire.

Water consumption depends on how the water is applied during the fight. When water is used to create a boundary line, it uses about 1 liter/m². When directly attacking the fire belt, it uses 3–5 liters/m², which corresponds to a rainfall of 3–5 mm. However, note that the applied water evaporates over time. Water consumption is also related to the type of fuel available in the area.

Smaller hoses can be used to minimize the amount of water that needs to be filled into the hose system and to reduce the workload for emergency personnel. However, if you are unable to obtain sufficient water, you should change the hose dimensions to a larger dimension. However, this means that more water is used to fill the hose system.

7.1.2 Extinguishing agent additives

Foam liquid admixture can lead to negative environmental impacts on both watercourses and vegetation and is therefore not recommended for use in forest fire fighting.

Long-term fire retardants are a collective name for substances that are added to materials to delay or suppress the progress of a fire. They are a mixture of water and salts that aim to chemically slow down the spread of fire in forests and soil. Long-term fire retardants have been used for several years, including in southern Europe, North America and Australia. The salts added to the water are intended to cool the fuel, change the pyrolysis process, lower the heat of combustion of the pyrolysis products and affect diffusion and oxidation in the reaction zone. The slowing effects persist even after the water has evaporated and long-term fire retardants applied to terrain sections in front of the fire can have a slowing effect on the spread of fire for more than 18 hours after application⁹.

9. Gould JS (2000). *Assessment of the effectiveness and environmental risk of the use of retardants to assist in wildfire control in Victoria*. Fire Management, Dept. of Natural Resources & Environment, Melbourne.

The application itself can be done either from the air or on the ground and is then done in connection with indirect attacks. The active ingredient in the long-acting fire retardants is ammonium salts, which in high concentrations have a negative impact on the environment. Chemical analyses of fire retardants used in the EU have shown the presence of cyanide, which has toxic properties¹⁰.

7.2 Brandgator

One method of containment lines involves emergency personnel separating the fuel from the fire using a firebreak. Typically, a highly makeshift, very narrow firebreak is started, with the most rapidly combustible material removed and taller fuels – such as bushes and tree branches – taken down to temporarily stop the spread of the fire. As time permits, the firebreak can be improved and, most importantly, widened to become a full-fledged firebreak.

When constructing a fire road in the terrain, personnel with chainsaws should take the lead in cutting, removing branches, clearing, etc., and carving the route. Following personnel work to remove the combustible material on the ground with various types of hand tools. Do not mix materials: burned material should go into the fire area and green material should go into the unburned part. Forestry machines or excavators can be used to advantage because the work with fire roads is heavy and demanding.

It is especially important to utilize existing fire barriers, to avoid making wide, labor-intensive fire streets. The construction should always start at an anchor point, go around the fire on both sides and end at the fire front.

The width of the fire escape is adjusted according to wind speed, slope gradient, fuel and fire effect development. A rule of thumb is that the fire escape should be at least 1.5 times the height of the surrounding vegetation. In extreme fire conditions, the fire escape should be more than 2 times the height of the vegetation. Be prepared to increase the width of the fire escape further if wind speed and flame height increase. These rules of thumb apply to a fire escape that must withstand the fire front and wings. Along the flanks and back, rescue personnel can reduce the width of the fire escape; in many cases, significantly narrower fire escapes can suffice in these cases.

To improve the effectiveness of the firebreak, water is applied to both sides of it. Fertilizer spreaders can be of great help here. However, water on both sides of the fire front also creates a diffuse and later difficult-to-control fire boundary and a risk of re-ignition than allowing the fire to advance to a non-combustible distinct area such as the mineral soil. Protective burning can also be carried out, and this method can be used to advantage since very large, labor-intensive widths of the firebreaks would otherwise be required (protective burning should not wet the fuel on the fire side).

NOTE! The entire width of the firebreak does not need to go down to the mineral soil; a few decimetres to one metre is usually sufficient. The part that goes down to the mineral soil is intended to prevent any smouldering fires in the soil layer from getting past the firebreak.

¹⁰Kärrman A (2021). *Knowledge compilation of environmental effects of long term fire retardants*. Report 1758, MSB, Karlstad.

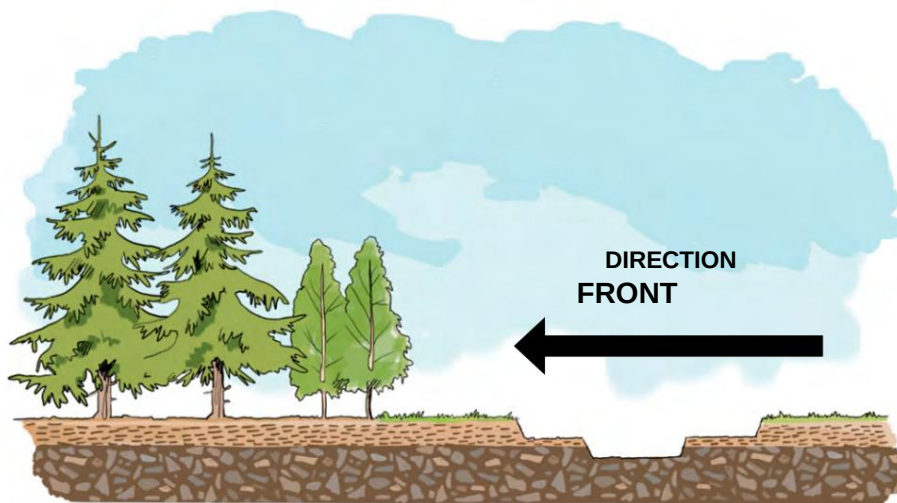


Figure 10. The structure and appearance of Brandgatan. Illustration: Christina Jonsson.

The excavated mineral soil can be used to retard the spread of the fire. The mineral soil is then thrown towards the base of the fire with a shovel. The mineral soil can also be used to cover fuel of a more compact nature near the fire path, for example stumps and logs.

NOTE: Covered stumps, logs, etc. must be uncovered once the fire has been contained to ensure that nothing is smoldering beneath the ground cover. Avoid leaving unburned areas within the boundary lines. Burn these areas when you determine it is safe to do so.

Make sure that the fire route also includes any aircraft fires that have occurred in front of the fire. However, avoid encircling the aircraft fires with separate fire routes, as this takes more time. The work will also be more difficult to monitor because personnel will be between two fires, which is a safety risk.

When constructing the firebreak, efforts should be made to prevent the forest fire from reaching areas of vegetation with lower moisture content and where there are larger amounts of fuel. This can prevent flare-ups and a worsening fire progression. Remove lower hanging branches from nearby trees to prevent them from igniting the branches, so that the fire spreads to the treetops. Introduce burning and glowing material into the fire area. Any fuel that cannot be removed when the firebreak is prepared for indirect attack should be placed on the opposite side of the firebreak as seen from the fire. The fuel on the opposite side from where the fire comes should also be wetted with water.

If protective burning is to take place, however, the fuel may be placed on the same side as the fire and should not be wetted. When the fire road is clear and the fire is approaching, there should be rescue service personnel deployed along the fire road for monitoring. If the fire jumps over the fire road, the personnel should be prepared to intervene. It is advisable to appoint special scouts whose task is to monitor any jumps over the fire road or aerial fires. Flying resources – preferably supplemented with a thermal camera – can be a good tool for detecting any jumps or aerial fires.

7.3 Protective burnout

The purpose of protective burning is to widen the boundary lines and secure them. It is done at the outer edges of the boundary lines on the fire side. When performing a protective burn, you should always start and proceed from a point that is considered safe to avoid the fire going around the protective burn, usually a fire road or other fire barrier.

It is also important that the protective burn is completed and extinguished before the forest fire front reaches the scene, as all emergency personnel should be focused on the approaching forest fire and not on the protective burn as the fire front approaches.

Protective burning is usually used to combat forest fires with high power development and high spread rate or in the event of an indirect attack. The chance of success with a protective burn is greatest with lighter and more homogeneous fuel. Take advantage of the daily variations in relative humidity when choosing the time for burning. When burning with heavier fuel, the risk of flying fires increases and the time it takes for the fire to burn out increases, requiring increased extinguishing resources. With varying types of vegetation, the risk increases that not all vegetation burns completely, which can mean that these terrain sections are instead ignited by the forest fire, which can then get past the burn and the boundary line.

The chance of a successful protective burn decreases with increased wind speed, changing wind direction and reduced humidity (the risk of aerial fires increases). In situations where unstable air masses exist, a protective burn can provoke the forest fire and result in an unpredictable fire course. This can then result in fire whirls and violent rushes of the fire.

Protective burning places great demands on preparation, organization and coordination. Safety must have the highest priority. A burn should not be carried out unless there are sufficient extinguishing resources to keep it under control.

A must during a protective burn is good communication. This of course includes the personnel carrying out the burn, but also that there is established radio or telephone contact with neighboring units, who can then be warned of any worsening fire progress, if the burn has jumped over the intended boundary line, etc. Neighboring units should be informed in advance that burning is to be carried out in the area and of the time when the burn will start. This is because the progress of the fire and the extinguishing work for neighboring units may be affected.

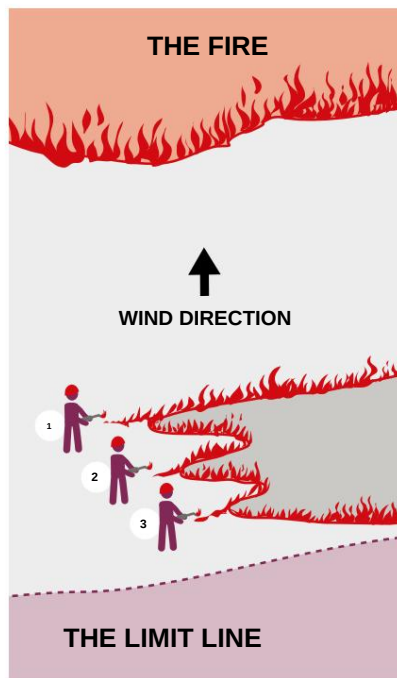
A fire officer should be appointed who is responsible for the burning operation and safety, and who does not participate in the burning itself but concentrates on overall supervision.

It is appropriate that this operation be a separate sector. The task is to continuously monitor both the fire and the personnel involved, to avoid anyone becoming trapped by fire.

To avoid exposing ground personnel to unnecessary risks, a helicopter can be deployed in these operations, both as backup for a quick extinguishing operation on a critical section of terrain and to light the sections of terrain closest to the forest fire with a so-called helitorch.

7.3.1 Ignition methods

Four ignition methods are described below. In all types of protective burning, involved emergency services personnel or other burning personnel should be informed in advance of the location of the nearest retreat point. It is also important that coordination and communication work well between the emergency services personnel or burning personnel who are burning.

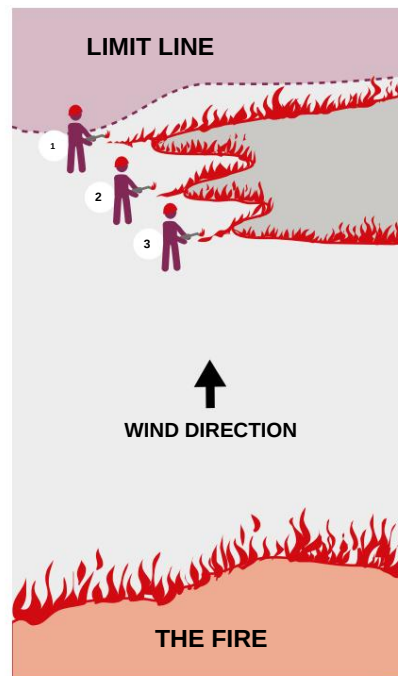


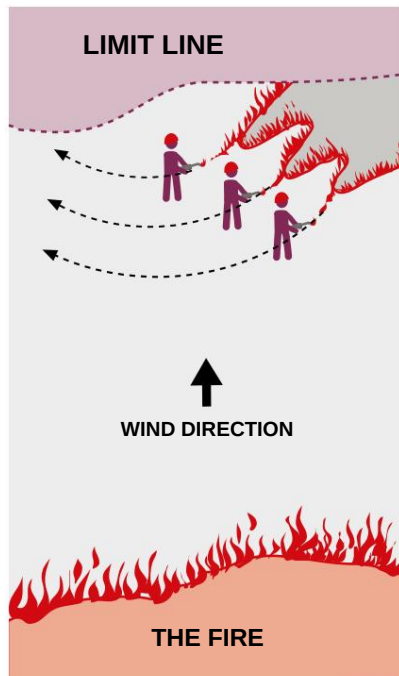
The first method is used when it is windy against the fire from the boundary lines. The personnel start from an anchor point, i.e. a starting point that is also an existing fire barrier such as a road, a mountainside or a watercourse, and ignite the vegetation closest to the boundary line. If something goes wrong, those who are burning should immediately head to the boundary line, which is then considered a place of retreat.

Figure 11. Protective burning when the wind is blowing against the fire from the boundary lines. Note that the person lighting No. 1 goes first.

When the wind blows from the fire towards the boundary lines, the next method is used. The first firefighter or burning person starts from the anchor point and ignites the vegetation closest to the boundary line, then the next person comes and ignites the area outside the already burned area.

Figure 12. Protective burning when the wind is blowing against the boundary lines. Note that the person lighting No. 1 goes first.





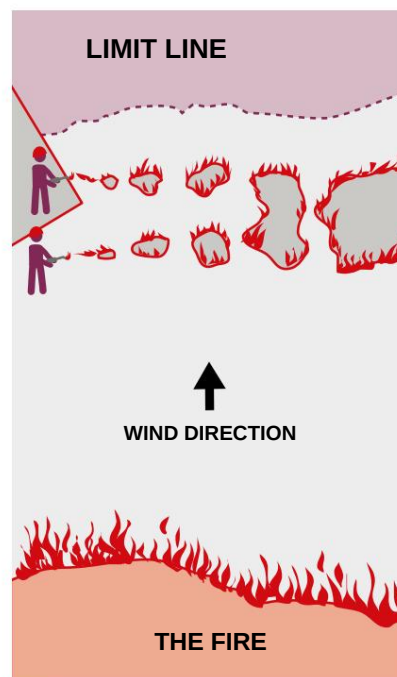
The third method involves lighting in a semi-circular motion and burning should then be carried out with the wind in your face and down any slope.

Firefighters or burn personnel start from the containment line. The method aims to pinch the fire front and is most useful in the areas around the flanks of the fire.

Figure 13. Ignition in a semicircular motion.

The fourth method is called spot ignition. The firefighters or burning personnel light in a spot pattern along the terrain bordering the boundary line and after a while the spots burn together.

Figure 14. Spot ignition.



7.4 Other methods

Other methods can also be applied, such as bog trampling, where the fuel can be pressed down into the water on bogs using, for example, a tracked vehicle. The same measure can also be effective in dry conditions, as the tracked vehicle then makes the fuel more compact and less flammable.

In the case of peat fires, so-called extinguishing sticks can also be used to access the deep-lying fire.

In the case of a low-intensity forest fire in lighter fuel such as grass, a leaf blower can be used to extinguish the fire. Make sure that the fire area is secured. The method is only suitable for fires with lower flame heights and when used, the direction of the air flow must be taken into account at all times so that, for example, glowing material is not pushed in an unwanted direction.

7.5 Flame length and choice of suppression method and containment line

The flame length can be used to assess what measures are required to stop the fire.

When the flame length is less than 1 meter, it is usually considered that a direct attack on the fire front or on the wings can be carried out. A simple, hand-made fire escape will then suffice.

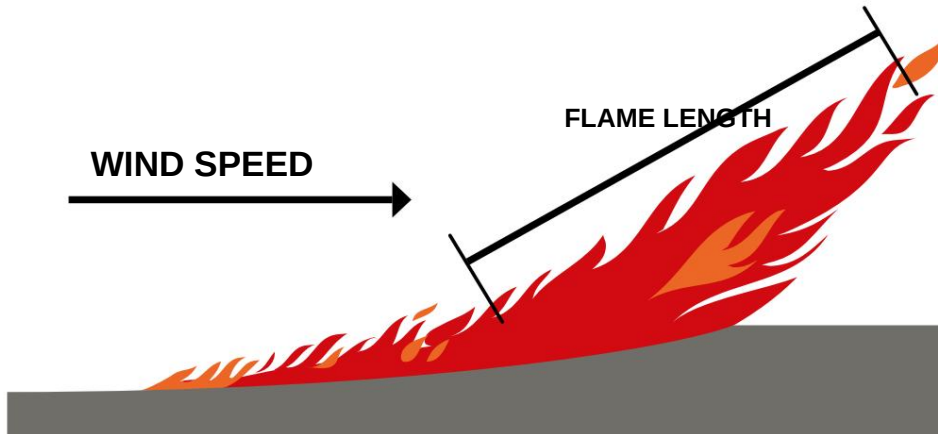


Figure 15. Flame length.

At a flame length of 1–2.5 meters, the fire becomes too intense for a direct attack on the fire front with hand equipment. It is also doubtful whether a temporary fire road will last. Additional resources may then be needed, such as harvesters, excavators, water-bombing helicopters, etc.

A flame length of 2.6 meters is the upper limit for using hose systems.

When the flame length reaches 2.5–3.5 meters, extreme forest fire conditions begin to occur. With high running fires, top fires and flying fires. Attempts to extinguish the fire by directly attacking the fire front itself are then probably in vain.

With a flame length of more than 3.5 meters, all extinguishing attempts against the fire will be in vain, but you can still make extinguishing efforts against the back, the flanks and possibly against the wings.

7.6 MSB's reinforcement resource with forest fire stations and high-capacity pumps

When the municipality's and region's own resources are not sufficient in the event of an accident, crisis or other major event, MSB has reinforcement resources in various areas that can be made available.

Municipalities should, as far as possible, be able to manage their own risks.

They are expected to collaborate within the region on events they cannot handle on their own.

For situations when the incident is too complex, too large or too long-lasting for the region's combined resources to be sufficient, MSB has reinforcement resources in various areas that can be used.

For support in the event of forest fires, MSB has, among other things, forest fire depots with extinguishing equipment and high-capacity pumps for water transport that can also be used as a firefighting resource.

7.6.1 Forest fire depots

A forest fire depot consists of three containers, mainly containing extinguishing equipment. It does not come with any support personnel as the equipment is loaned out as the equipment is familiar to fire personnel.

The depots are kept at various rescue services around the country. There are a total of 24 depots, some of which are reserve depots in MSB's central warehouse in Kristinehamn.

A depot contains, among other things:

- 23 km of fire hose of various dimensions
- Jet and manifold pipes
- Motor sprayers, towable and portable
- Water sprinkler
- Hose reel
- Chainsaws
- Six-wheeler with trailer



Figure 16. Parts of a forest fire depot. Photo: Robert Hjalmarsson.



Figure 17. Parts of a forest fire depot. Photo: Robert Hjalmarsson.

7.6.2 High-capacity pumps for water transport and firefighting

MSB also has two high-capacity pumps with a capacity of 15 m³/min.

These pumps can transport water over long distances and are therefore suitable for both flood and forest fire operations. The pump resource has automated hose deployment and reeling.

When the high-capacity pumps are used, personnel accompany and manage their operation.

Night work

8. Night work

Night is usually ideal for fighting a forest fire because the intensity of the fire decreases during the night. Humidity is usually highest at night, the wind has subsided and the temperature has dropped, which is beneficial for the physically demanding firefighting work. With the decreasing fire intensity, fires at night can often be fought with a direct attack, which is very beneficial from a firefighting perspective. Night can also be used for, for example, improving existing boundary lines, performing maintenance on engine sprayers, etc.

Night also brings with it a number of changed circumstances that are important to consider. Fire smoke is not as easily observed visually as during the day. This means that any aerial fires that could have been detected during the day via the smoke column are usually detected via the flames. Fire-damaged trees that are at risk of falling may be more difficult to detect. The risk of sprains and falls also increases during the dark hours of the day. Consider only working along existing boundary lines if there is a high risk of personal injury.

There are a number of different measures to facilitate work at night. In addition to helmets, headlamps or flashlights, staff should also wear light reflective vests. Consider using a handheld thermal camera to facilitate the detection of fire sources. Hazards, paths, escape routes, etc. can be provided with reflective tape. If possible, night staff should have been in the same area during the day to increase orientation. For example, shift changes can take place before darkness falls. Topographic maps or orientation maps are excellent material for drawing attention to risk areas for fall accidents, but also for seeing where boundary lines can be laid, vehicles can move, etc. Night orientation is also an excellent exercise.



**Securing the
outer areas
of the fire**

9. Securing the outer areas of the fire

After the forest fire has been enclosed by containment lines and the further spread has been temporarily stopped, work begins to secure the outer areas of the fire closest to the containment lines. This is to ensure that the containment lines withstand sudden flare-ups and that no flying fires occur beyond the containment lines. In the case of smaller forest fires or low-intensity forest fires where the occurrence of smoldering fires is low, the entire fire area is usually extinguished completely without any major problems.

In the case of larger forest fires or high-intensity forest fires with a high incidence of smoldering fires, a complete extinguishing of the entire fire area becomes impossible due to the resource requirements and the long time aspect. In these cases, most of the work is focused on the area closest to the boundary lines. The extent of securing the outer areas is then determined to a very high degree by the weather and the fire risk forecast for the coming days and weeks.

As an absolute minimum, all hot spots within a distance of 20 meters from the boundary lines must be located and extinguished. In the event of unchanged or worsening weather and fire risk forecasts, the distance is increased to 40 meters¹¹.

Areas of terrain within the fire area that have not burned are either burned off or surrounded by a fire road. Trees with smoldering or open flame fires in the upper layers are felled. All heavier fuels in the burned area that can be removed, such as larger branches or logs, are moved at least 50 meters away from the boundary lines.

In areas in close proximity to buildings or cultural and historical environments, the above distances will need to be further extended.

The area closest to the boundary lines can be advantageously divided into sectors to increase the systematic nature of the work.

The rest of the fire area can be checked more sporadically, for example by flying over with a thermal camera or by patrolling through the fire area, where a handheld thermal camera can be used to advantage, and the focus is placed only on probable locations for fire sources. However, wind-exposed parts within the fire area itself should be checked and smoldering fires should be located and extinguished. Other resources such as smoldering fire dogs can also be used to locate smoldering fires.

11. Chandler C. et al., *Fire in Forestry, volume 2, Forest Fire Management and Organization*, Krieger Publishing Company, 1991.

9.1 Practical advice

Below are tips and advice for securing the outer areas of the fire:

- The actual work on the fire surface must be characterized by precision.
Reignition must not occur.
- Start with the area closest to the boundary line or risk objects that are considered to be capable of causing fire spread across the boundary line, for example branches, bushes, terrain areas, etc. that are burning in close proximity to the boundary line.
- Completely or partially burned material is advantageously thrown further into the fire area.
Black material should enter the black area (i.e. burned area).
- Fuel on the opposite side of the boundary line may also require some measures to minimize the risk of ignition across the boundary line. Accumulations of fuel are removed from the area closest to the boundary line. Green material should enter the green area (i.e. unburned area).
- Use a fine, diffused jet and sweep over a smaller area.
fire surface, this is to detect hot spots where the water is evaporating. Partly by listening for hissing sounds, and partly visually by observing any water vapor.
- Also watch out for smoke and swarms of mosquitoes, which are attracted to heat.
- Watch out for fire-damaged trees that can suddenly fall. Leave them standing as they play a major role in wildlife in the forest. However, if they pose a danger or if there is a smoldering or blazing fire in them, it will of course be necessary to fell them.
- Break up larger fire-damaged fuel pieces with hand tools to cool them more effectively.
Chop up and break up embers.
- Many times it can be enough for material to be moved around, spread, chopped opened or exposed because the increasing humidity at night can then extinguish the embers.
- Be observant and check carefully for anthills, as these are difficult to extinguish.
- Make sure to turn logs and the like over as embers can be hidden on the underside.
- Feel with your fingers where embers still exist. However, watch out for hot stones. If white ash is visible on the surface of the stone, there are often hot embers and high heat underneath. Do not bury embers without later digging them up again and making sure they are extinguished.
- Work in pairs. One digs and disperses while the other cools and extinguishes with water. Turn off the jet pipe at regular intervals to check the power and effectiveness.
A mini excavator can make the job easier.
- Areas with greater depth of soil layer are either dug out and spread or soaked heavily with, for example, a fire extinguisher and mixed/stirred.
- Pay attention to roots and stumps! Consider the risk of spreading underground.
Dig up the stump or roots and feel for heat if you are unsure.

- IR/thermal cameras can be used to find fire and embers both on the ground and from the air. Carry out this scan early in the day or late in the evening to avoid sources of error such as rocks heated by sunlight. Keep in mind that the thermal camera only “sees” the outer surfaces, which means that a fire source behind a tree is not registered by the thermal camera, only the temperature of the tree is registered. The sun can heat up rocks and other objects that the thermal camera registers as hot objects. Therefore, the rescue service is best able to use thermal cameras at sunrise or a few hours after sunset, before and after the sun has set.

9.2 Monitoring

Surveillance of the fire area can last for a long time – several days or weeks. The number of rounds is reduced as the days go by and the number of detected fire sources decreases. A recommendation is that at least two daily rounds are carried out on foot and that one of the daily rounds is carried out at 3 pm in the afternoon, since the risk of possible flare-ups is greatest then.

Monitoring may also need to be carried out more frequently when the weather changes to more unfavorable conditions, for example strong winds.



Termination of rescue services

10. Termination of rescue services

Below are tips and advice regarding practical aspects in connection with the termination of emergency services:

- Initiate cooperation with landowners as early as possible in the operation. They are a good resource in the extinguishing work and they will then have time to familiarize themselves with the job before the end of the rescue service.
- If individual forest owners agree, coordination can, with advantage, take place, for example, through a designated spokesperson, forest owners' associations, companies that feel responsible towards landowners through previously made deals or other means. •

NOTE: Residual value managers as coordinators can only be relevant if the forest is insured, which in many cases it is not.

Initiate regular landowner meetings early to coordinate collaboration, report on the situation and plan for continued operations.

- Make a forecast for when the termination of rescue services may be necessary and communicate this with the landowners so they have time to prepare.
- Before closing, check that the landowners are ready and assess whether they are capable of completing the task: access to firefighting equipment, personnel, helicopter, etc.
- How long the rescue service should last is based on the four criteria for rescue service according to the LSO. Linked to the criterion "Other circumstances", the conditions for terminating the rescue service may vary depending on whether it is an individual landowner with weak resources compared to a large forestry company. Make an assessment on a case-by-case basis.
- If there are large areas and many sectors, it is possible to end the operation sector by sector. Rescue operations will then continue in the remaining sectors. If there is a re-ignition in a sector where rescue services have ended, this means, legally, that a new rescue operation will be started.
- Explain to landowners what is expected of them after completion of rescue service. The description can be detailed, but at the same time it must be clearly emphasized that control intervals, staffing levels and the like may need to be reconsidered in the event of changed weather conditions, etc.
- Clarify for the landowner what problems exist within each sector: water supply, hotspots, possible weaknesses in boundary lines, etc.
- Make a **written** conclusion of the rescue service.



Combat from the air

11. Air combat

11.1 Request for Airborne Support

When the fire service is faced with forest fires with rapid spread, extensive fire growth and when their own resources are deemed insufficient, MSB can offer various types of reinforcement resources, such as airborne support with helicopters and scooping aircraft.

MSB's reinforcement resource for flying support in the event of a forest fire aims to support and strengthen the ability of the responsible actors to manage a forest fire and thereby limit its spread.

If the forest fire is assessed to be likely to develop rapidly and extensively and if airborne firefighting resources can be vital in reducing the progress, resources should be requested early.

MSB's flying resources should be used primarily if no other local or regional resources are available that can begin an operation more quickly.

If MSB's flying resources are not sufficient, there is the possibility of requesting the Armed Forces' resources and requesting support with international resources via MSB.

11.2 MSB's reinforcement resource by helicopter or airplane for forest fire extinguishing

During the 2022 forest fire season, there will be access to water-bombing helicopters and bucketing aircraft with crew and equipment for forest firefighting.

The helicopter resources are normally spread out over various bases in the country, the positioning is adapted to the risk picture. They are self-sufficient in terms of fuel and carry equipment and personnel for communication with rescue leaders. If necessary, MSB can decide on increased readiness with the helicopters based on a combination of the fire risk, any Ahotade values and the need for resources. Readiness can then be increased through shorter stress times – i.e. changing the readiness level – and more helicopters on standby.

The helicopters can fly around the clock if weather and light conditions permit.

The scooping aircraft have a capacity to release up to 3,000 liters of water per drop. The aircraft resource includes crew and necessary associated logistics.

The aircraft resource is normally located at Skavsta (Nyköping), but depending on the risk situation, the positioning can be changed. As with the helicopter resource, the readiness of the aircraft can be increased by changing the readiness level.

During the forest fire season, there is a readiness of at least ten helicopters and four bucketing aircraft. Two of the bucketing aircraft are on EU readiness from April to October. Helicopters can also be on EU readiness during certain periods if Sweden and the EU conclude such an agreement. EU readiness means that the EU finances a large proportion of the costs of the readiness. The resources on EU readiness can be used for national efforts. Any priorities are made in dialogue between Sweden (MSB) and the EU. Through cooperation within the EU Civil Protection Mechanism, good opportunities are created to both give and receive support. This strengthens both Sweden's preparedness and the EU's overall preparedness in the forest fire area.

An authorized decision-maker within an emergency services management system has the possibility to request reinforcement resources from MSB according to LSO, via the authority's emergency officer, TiB. If reinforcement resources are needed, contact MSB's emergency officer, TiB, by telephone: 054-150 150. The call is connected via SOS Alarm. E-mail: tib@msb.se.

After initial contact and when possible, the request is completed on a special form (see Appendix 1). Please note that the form should be considered dynamic and may change in design during the forest fire season.

When airborne support is requested with reference to LSO, the MSB is responsible for the costs.

Communication between the person/role leading the rescue operation and the pilot can primarily be via RAKEL, but in some cases a mobile phone or VHF aircraft radio may be an alternative. Which of the alternatives will apply is clarified when the resource is activated.

Please note that there are both aviation VHF and marine VHF where the frequency band differs between the two cases. In this guidance, VHF radios equipped with aviation frequencies are referred to.

Communication is an area where development work is ongoing, if changes occur during 2022, these will be announced.

11.2.1 Preparedness levels

Helicopter on standby

Helicopters on standby must take off within 90 minutes after MSB alerts the resource.

Aircraft

Normal tensioning time is a maximum of 180 minutes.

Shortened tensioning time is a maximum of 30 minutes.

11.3 The Armed Forces' helicopter resources

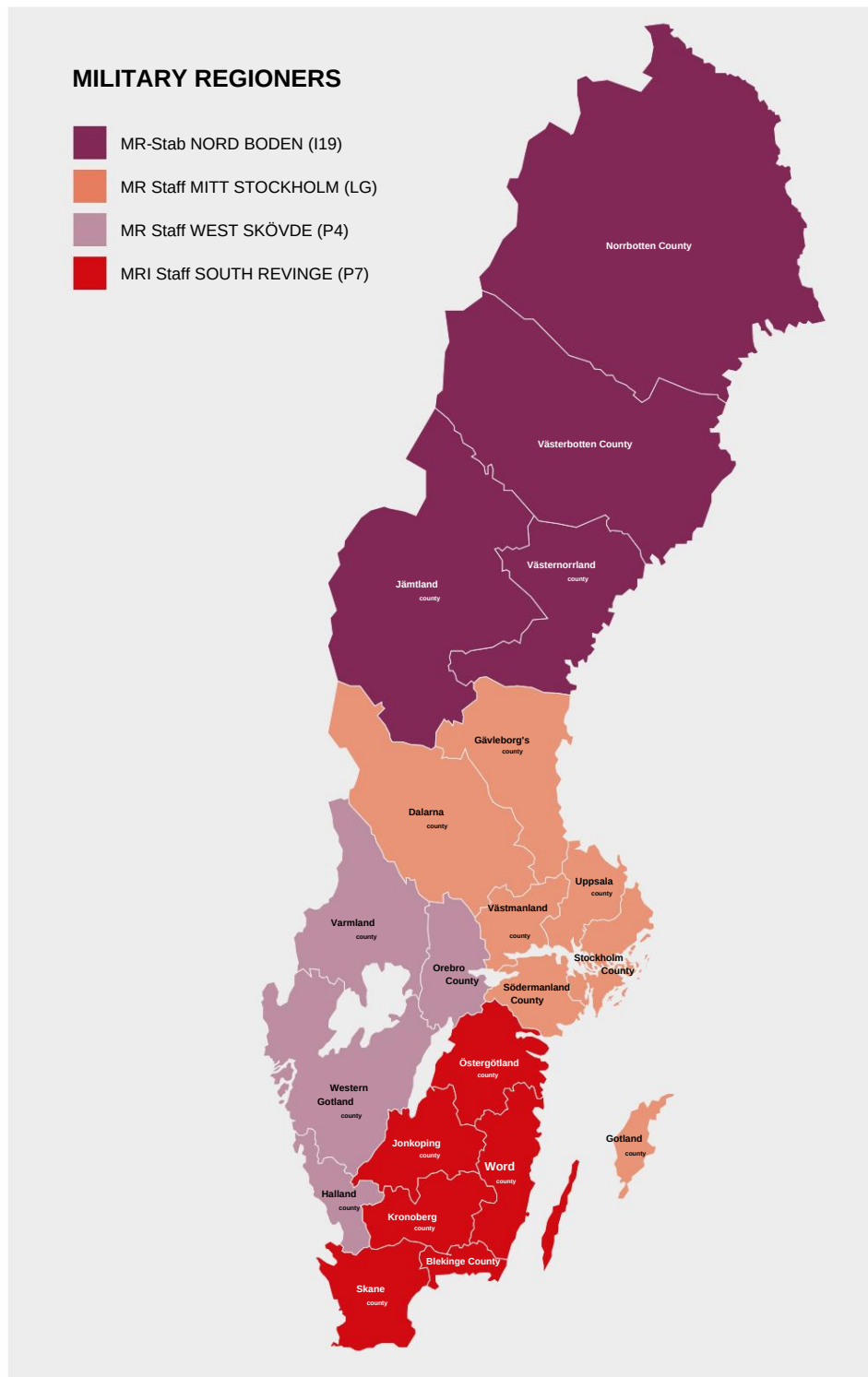


Figure 18. The military regions of the Swedish Armed Forces.

The rescue leader makes a request for support with the Swedish Armed Forces' helicopter resources (according to LSO) to the relevant military region. The rescue leader first makes verbal contact and then makes the request on a special form (see Appendix 2 and Appendix 3).

Private contact can be made with the officer on duty at the Helicopter Flotilla if support is needed, for example, regarding what a requisition should include and so that measures can be prepared for a decision.

The Commander-in-Chief (VO) of the Swedish Armed Forces at Headquarters makes decisions on operations with the Swedish Armed Forces' helicopters after preparation with the Air Staff and the Helicopter Flotilla.

The helicopter flotilla can contribute the following capabilities to fighting forest fires from the air:

- Medium-duty helicopters with 2–2.5 cubic meter water barrel.
- Light helicopters for reconnaissance, equipped with, among other things: thermal camera.
- Management and liaison support to Rescue leader for coordination and management of the operation in the air.
- Logistics needed for aviation operations, such as fuel vehicles.

In a helicopter operation, a command and control resource should be brought along for communication and for effective command and coordination of the operation. The need for command and control capacity grows as more helicopters participate in the operation. The connection to the helicopters is via frequencies within VHF/

The UHF band (not Rakel). However, the command resource has radio systems for both Rakel and communication with the helicopters.

The command capability can also be requested separately even if the Swedish Armed Forces' helicopters are not participating in the operation.

A request for support should describe in detail which capabilities you want to have access to in order to avoid interpretations or misunderstandings.

MR Nord, VB

Email: ~~vb-mrn@mil.se~~
Phone: 0921-34 91 00
Fax: 0921-34 84 45

MR My VB

Email: ~~mrn@mil.se~~
Phone: 08-58 45 40 42
Fax: 08-58 45 40 80

MR West VB

Email: ~~vb-mrv@mil.se~~
Phone: 0500-46 61 00
Fax: 0500-46 61 01

MR South VB

Email: ~~vb-mrs@mil.se~~
Phone: 046-36 88 00
Fax: 046-36 80 50

Helicopter Flotilla World Championship

Email: ~~vb-hkpfli@mil.se~~
Phone: 013-28 32 50

11.4 Coast Guard Aviation: Sensors and Conductivity

The Coast Guard has three similar aircraft, all based at Skavsta outside Nyköping. The crew consists of four people: two pilots and two system operators. The endurance in the air is 6–7 hours and the working hours for the crew are up to 14 hours per day. The Coast Guard aircraft are not on standby 24 hours a day, all year round. However, at least one crew is on duty every day.



Figure 19. Coast Guard aircraft, model DASH 8 – Q300. Photo: Coast Guard.

The aircraft are equipped with a range of different sensors that can be used for, among other things, mapping a forest fire:

- EO (electro-optics), which includes thermal imaging (IR and SWIR), as well as daylight cameras. The system generates video and still images with high positional accuracy.
- IR/UV scanner where the IR scanner can be used for mapping the ground surface.
- Handheld still camera for high-resolution images with positioning.

The following communication options are available:

- 2 VHF aviation frequencies.
- 2 VHF maritime channels.
- 2 st HF + 1 VHUF.
- 2 satellite phones.
- 4G datalink.
- SAR direction finder, ability to find different frequencies.
- 2 RAKEL (NatBlue with the possibility of linking with e.g. RAPS).
- Repeater and Gateway function (RAKEL).
- AIS / TCAS (to see other flying units).
- Mail (possibility to email and MMS individual phones).
- Possibility to export SiTaC symbols and polygons in KML/KMZ format (compatible with Google Earth, Daedalos, Oculus etc.).

The Coast Guard flight can assist with:

- Location pictures/maps.
- Documentation of areas and properties.
- Relay radio traffic.
- Directly manage and distribute resources, e.g. fire engines.
- ACO function for flying units.
- Establish R-areas, i.e. restriction areas.
- Workplace for a rescue leader, fire officer or equivalent.
- Investigation, evaluation, research.
- In the case of longer operations, the possibility of providing the staff with its own personnel for to handle information and flight issues.

11.4.1 Airspace Coordination, ACO (Air Coordinator)

The Coast Guard has a very good ability to lead other flying resources within a given area. This is trained and practiced annually with a focus on leading rescue helicopters over seas. Everything then takes place according to a global concept that is described in more detail in IAMSAR vol 3. The purpose of this training is to achieve the highest flight safety and an efficient flow of helicopters around a major casualty (ex Estonia).

The Coast Guard's ability to perform ACO has proven very useful. especially during the major forest fires at Sala in 2014 and Ljusdal in 2018.

For airspace coordination, frequency 123.1 MHz is used according to standard procedure. Otherwise, a different frequency will be agreed upon.

Management according to the ACO concept requires an aviation background and a great deal of training.

Due to, for example, obstructing terrain, ACO is advantageously carried out from the air.

The need for ACO should be considered early if it is realized that the operation will be extensive and where a larger number of helicopters or a mix of both helicopters and aircraft is expected. Individual rescue leaders can have a dialogue with the flying units on site and also with the Coast Guard Flight to assess the need for ACO.

Contact information Coast Guard Flight:

Management center in Gothenburg, accessible 24/7

Officer on duty

Phone: 0776-70 60 00

E-mail: Lc@kustbevakningen.se

11.5 International resources

If national resources are insufficient, support from other countries may be requested.

We can do this through various agreements and collaborations. We can, among other things, turn to our Nordic neighbors for support. Through the Nordred agreement, a rescue leader can contact actors in our Nordic neighbors and ask for support in a rescue service situation.

There is also a system within the EU's civil protection cooperation, the Civil Protection Mechanism, which is prepared so that countries can smoothly support each other with resources. MSB is here the national contact point towards the EU. If a rescue service wants support with resources via the EU, it therefore contacts MSB's TiB. MSB assesses the reasonableness of the request and makes the possible request for resources. MSB shall also inform the government when such a request is made to the EU.

The actor requesting the support has the formal responsibility for receiving it in a good way. However, MSB has a special reinforcement resource, Host Country Support, which can assist in receiving and coordinating international support. This resource is also requested through MSB's TiB.

MSB's *Guidance on Receiving International Support* (ISBN 978-91-7383-608-1) describes the collaborations available for international support and what needs to be handled when requesting, receiving and coordinating support. In 2022, it will be supplemented with an appendix and checklist on host country support when requesting support with flying firefighting resources.

11.6 Helicopter use

The helicopter is a very flexible tool for fighting forest fires and can be used for a variety of tasks, including:

- water bombing
- water transport (filling a pool or tank in the terrain on hard-to-reach places)
- transportation of personnel and equipment
- rescue of personnel
- forest fire reconnaissance
- ignition and protective burning of areas with the aim of creating fire streets using, for example, helitorch.

The flying resources supplement the ground resources in difficult-to-access terrain, in fire areas with high power development where a direct attack from ground personnel is not possible, and in the case of aerial fires.

The use of flying resources in a forest fire can be divided depending on when the resource is deployed. The flying resource is most useful in connection with an initial intervention. The resource is then used quickly to prevent the fire from increasing in size, becoming difficult to extinguish and resource-intensive. The flying resource's task is then, among other things, to keep the fire down until the emergency services have resources on site and the emergency services personnel can gain control of the fire.

A rapid initial response with flying resources is particularly important in areas with sparse road networks and fewer emergency service resources. Periods with extreme fire risk values are also examples of when an initial response with flying resources is extremely important to quickly contain the fire.

In the event of major forest fires, the task of the flying resources is changed to instead:

- carry out supporting water bombing to delay the spread of the fire dispersion in certain areas in order to create time for ground personnel to prepare containment lines
- carry out water bombing to reduce the fire intensity and thus, by minimizing risk, enable ground forces to enter the area and begin a direct attack
- protect buildings and carry out targeted interventions in areas where the risks are high for the rescue services

constitute a first response resource after a thunderstorm has passed over the fire area.

Water bombings are carried out in collaboration with emergency services personnel, where emergency services personnel enter the fire zone as soon as the fire suppression has been carried out to take advantage of the dampening effect that the suppression has on the fire intensity. Sectors should not be emptied of emergency services personnel just to carry out water bombings, but the safety distance to the fire zone in connection with the water bombing must of course be observed.

In the case of larger forest fires, a flying resource with a greater extinguishing capacity is often of greater importance than flying resources with a smaller extinguishing capacity. This is to increase the chance that the water will actually reach the desired location, to cover a larger area or to obtain a larger amount of extinguishing water per unit area with each water bombardment. However, lighter helicopters can also play an important role in the firefighting of a larger forest fire, partly to obtain continuity in the application of water, and partly in the form of a first response resource against emerging aerial fires.

NOTE: Flying resources should be seen as a complement to ground resources; a forest fire can be extinguished by ground personnel alone, but never by flying resources alone.

11.7 Leading a single helicopter

Communication with the helicopter should be arranged as early as possible. RAKEL should be used as a means of communication in the first place, but mobile phones can also be an option. However, all helicopters are equipped with VHF radios, and another possible solution is for the helicopter to land and hand over a VHF radio to the rescue service personnel. If an extra VHF radio is not available, it must be arranged how else to communicate.

It is advantageous to photograph and send a situational picture with SiTaC symbols (see section 13.3) to the crew at the first contact with the helicopter crew. The crew gets an idea from the start of where the commander leading the operation at the scene wants the helicopter to be deployed and where ground units, command post, critical points, etc. are located.

Once the helicopter has arrived at the fire scene, the rescue leader or another fire officer will accompany you into the air for an initial reconnaissance of the scene. The work on the situation picture continues in the air where plotting is done on a simple map (preferably with SiTaC symbols) for example using a regular camera or a thermal camera.

If necessary, the relevant airspace can be closed after a decision by the rescue leader. Applications for restrictions on aviation are sent to the Swedish Transport Agency.

After the tactics and direction of the water bombing have been decided – with advantage in consultation with the flying crew – the actual water bombing begins.

Take into account any restrictions regarding watercourses from which to collect firefighting water and that the same barrel should not collect water from several lakes due to the risk of spreading crayfish plague. The decision to collect firefighting water from a private pond must be made by the rescue leader and documented, as this involves encroaching on another person's rights. Water bombing is carried out under the direction of the rescue service personnel in close cooperation with the flying crew, and here too communication plays a major role. The rescue service personnel should provide feedback to flying units about the effects of the water bombings carried out, as the helicopter crew rarely sees the full effect of the water bombing. The helicopter crew should communicate any flare-ups, aircraft fires that have occurred or impending water bombings to the rescue service personnel on the ground.

In some cases, rescue service personnel may direct a helicopter to a specific point in the terrain. This can be done in various ways, where a developed map with a grid system can be used as a basis. One method is to direct the helicopter using characteristic landforms or vegetation, for example, "Northeast of the big rock, distance 150 meters". The characteristic landform can also be called a "bullseye". Another method is the clock method. Straight ahead, the helicopter crew has 12 o'clock, straight back has 6 o'clock, to the left has 9 o'clock, etc. The helicopter is then directed by saying, for example, "Nine o'clock, distance 200 meters". A third method is for rescue service personnel to mark out the places where they want the water bombing to take place, for example with brightly colored markers.

11.8 Waterbombing tactics

Helicopter water bombing tactics have some similarities to the tactics used by ground rescue services, including using direct and indirect attacks. However, the tactics also differ due to the different tasks.

The flying resources are mainly used for spot operations with the aim of creating time for the fire service, reducing the fire intensity and enabling ground operations.

This can be compared to the effort on the ground which aims to contain the fire, stop further fire spread, ensure that the areas closest to the boundary lines are extinguished, etc.

If possible, the flying resources should be concentrated during the initial stage of the fire. This is to avoid tying up flying resources to a possible fire for a longer period of time. It is better to concentrate early, get an early decision and detach the flying resources to be prepared for a new alarm.

During the initial stage, one should avoid using both helicopters and airplanes for water bombing as a mix of flying resources requires an organization that can take time to get in place.

11.8.1 Direct and indirect attacks

Direct attack means that the water bombardment is carried out directly on the fire band or in the immediate vicinity of the fire band and then in close cooperation with the rescue service personnel on the ground. For example, half the load can be dropped on the fire itself and the other half in front of the fire, this to achieve a speed and intensity reduction of the fire. Direct attack is used in frontal attacks, flank attacks, aerial fires and sudden flare-ups (see more about this below).

Indirect attack means that the felling is carried out on the fuel that is further away from the fire band in the path of the forest fire, i.e. the fuel is wetted down, possibly in connection with fire streets.

In Sweden, in most cases, it will only be a direct attack due to the evaporation of the water and its decreasing effect over time. The text below will therefore focus mainly on the direct attack.

11.8.2 Frontal attack

Deposition at the fire front can reduce fire intensity and spread rate to enable emergency personnel to make a direct attack on the area.

In the event of excessively high fire intensities (flame heights >3.5 meters) or in the event of an excessively long fire front, a frontal attack is highly questionable. In the latter case, however, the fire front can be divided into smaller and more manageable sections through a number of concentrated fellings. The concentrated fellings should be made in areas or sections where the topography or vegetation is favorable and increases the chance of successful division of the fire front. In many cases, the frontal attack also includes the most critical or intensive wing to prevent the fire front from growing in size. In the case of an already extensive fire front, the effort can therefore be focused mainly on the most active wing to push the fire back and shrink the fire front in this way.

11.8.3 Flank attack

In the case of very large and intense forest fires, it may prove impossible to initially go to the front or wings themselves. The focus will then instead be to perform fellings along the flanks while waiting for the power development and the speed of spread to decrease. The aim of the fellings is then to finally work towards a wing and squeeze the fire front together. The crew may try to steer the fire front towards a fire barrier, such as a watercourse, in order to stop the spread of the fire.

11.8.4 Aircraft fires, hotspots and sudden flare-ups

One of the most important tasks for a helicopter during a major forest fire is to extinguish or suppress any aerial fires, hotspots and flare-ups.

This is important to prevent extreme forest fire events. The flying fires contribute significantly to the fire's impact, as they burn together with the main fire. If the fire starts to "spit", i.e. spread flying fires, it becomes a behavior that constantly escalates and feeds itself.

The helicopter's task is to contain the fire until emergency personnel arrive at the scene. The first cuts are made on the aerial fire itself and on the fuel just in front of the front. The following cuts are made on the rear parts of the fire and on the fuel around the fire itself.

After initial reconnaissance, the initial focus may be to invest on extinguishing or containing so-called hotspots – i.e. areas or points with high fire activity or high temperatures. Hotspots are identified by the color of the smoke, clear movements in the smoke and clear and distinct flames. As the operation progresses, aimlessly combating hotspots is not recommended because hotspots can continuously occur here and there in the fire area. In such cases, a better solution may be to coordinate with the ground units and gather forces in prioritized areas and to initiate fire suppression over the most intense hotspots.

11.9 Other influencing factors

11.9.1 Penetration of branches by precipitation

The drop from a helicopter is controlled by the direction, height and speed of the drop itself. The ability of helicopters to greatly vary the speed during water bombing means that drops from helicopters are significantly more flexible compared to drops from water bombing aircraft. For example, a stationary drop is an option when using helicopters but not when using aircraft.

When felling from a helicopter, the water concentration will be highest in the center of the felling area and decrease with increasing distance from the center. When water bombing with helicopters with high lift capacity, the area with high water concentration will be larger and therefore have a greater chance of penetrating through branches. The ability to penetrate through branches is particularly desirable in dense forest where the fire is at ground level.

11.9.2 Access to water

The key issues when assessing the application of extinguishing water are how continuous the application of extinguishing agent is and the amount of extinguishing agent that can be applied. Governing factors are the orbital time (i.e. the length of time between helicopter drops), the number of water-bombing helicopters, the lifting capacity of the helicopters and the possibility of refueling with aviation fuel in close proximity (to avoid interruptions in the water-bombing).

In the event of a forest fire where water application is too sporadic or low, measures may consist of increasing the number of helicopters or using heavier helicopters, using watercourses closer to the fire, or directing flying resources along less intensive sections such as the wing or flank.

In very dry conditions and extreme fire risk values, at least two flying units should be used to ensure effectiveness and continuity of extinguishing agent application.

Regarding possible cancellation of watercourses, see section 16.2.1.

11.9.3 The work of emergency service personnel

If the emergency services personnel are advancing quickly, the focus may be, for example, to constrict the fire front. If the personnel are advancing slowly, the focus may be to only contain the most critical sections of the fire.

It is important that rescue personnel do not back away too far from the fire line during aerial suppression. If the distance is too great, personnel cannot take advantage of the reduction in fire intensity after suppression, and the fire may then increase in intensity again, making direct attack by the rescue services impossible.

11.9.4 The smoke column

The direction and spread of the smoke column will determine the choice of tactics to a large extent due to visibility requirements and the fact that flying through smoke columns should be avoided. The smoke column can therefore significantly limit the possibilities and, for example, make a frontal attack impossible and mean that the operation is instead directed towards the flanks and wings.

Example 1:

A forest fire burns in relatively open vegetation (no complete tree cover) with a wind speed of 5 meters/second and a relatively narrow fire front.

The helicopter's primary task may then be to drop at/on the fire front to cool it down and reduce the rate of spread. Depending on where the ground crew enters the area, for example, the next drop may be focused on the most active and critical wing.

In the event of a forest fire with a longer fire front, or a fire front with very high power development, the approach might have been to first and foremost focus on the most critical wing and then work to reduce the length of the fire front.

Example 2:

A forest fire is burning with a narrow fire front and an aerial fire has arisen in front of the fire front. The initial task for the helicopter may be to carry out a frontal attack on the fire front to reduce the fire intensity and prevent further aerial fires. If several helicopter resources are on site, the aerial fire should also be kept down in parallel with the frontal attack. Finally, a frontal attack on the fire front and the wings can be carried out to compress the fire front.

If a frontal attack on the fire front is not possible, the primary task may instead be to hold down the aerial fire and to carry out a flank attack or possibly a frontal attack against the most critical wing.

11.10 Water-bombing aircraft

Scooping aircraft are a powerful tool in forest firefighting and can be used for multiple tasks.

The tactics when using water-bombing aircraft are largely identical to the tactics when using a helicopter.

However, there are a number of differences due to different technical conditions.

The bucket-type planer has a collection capacity of up to 3,000 liters of water per release.

The drop area can be both concentrated and have a more extended appearance depending on the tactical drop setting in the plane. However, the precision of the drop will generally not be as high as for a helicopter, which means that emergency service personnel on the ground must be prepared to retreat further during a drop compared to from a helicopter.

Stationary felling is also not possible, but instead the approach can be made in a downward trajectory to increase the possibility of penetrating branches. The requirements on the watercourses from which water can be scooped are also higher compared to a helicopter. This means that the orbit time is usually longer.

To ensure continuity in water application, this means that at least two aircraft operate at each forest fire and never a single aircraft. The distance to the nearest refueling point can be long, which may affect the continuity of water application.

11.11 Management of multiple helicopters and airplanes

If the number of water-bombing helicopters increases and if water-bombing aircraft arrive, the complexity and need for control of the flying resources increases, both in terms of safety and the effectiveness of the flying fire-fighting operation, communication, etc. In connection with this work, the rescue leader can also request support from MSB linked to staff and management support, but also support linked to liaison management as well as management of rescue operations at various levels.

Other actors are the Armed Forces and the Coast Guard, which can provide assistance with management resources and expert personnel.

11.12 Safety aspects when using a helicopter

It is important to consider your own safety as well as the safety of others when using a helicopter.

The danger zone around a helicopter is 5 meters outside the outer end of the main rotor and 5 meters around the tail rotor. The risk zone will therefore vary depending on the helicopter type. However, during takeoff and landing, the risk zone is 30 meters around the helicopter. If the risk zone must be entered, this must be done while maintaining eye contact with and following signals from the crew.



Figure 20. The danger zone is 5 meters outside the outer end of the rotors.
Illustration: Per Hardestam.

Keep the following in mind when approaching or within the risk area:

- Approach and move away from the helicopter so that you have eye contact with the pilot, and wait for signals before approaching.
- Approach the helicopter from the front or from the side, never from behind.
The most common approach is to approach the helicopter from the front at an angle. If you approach from behind, the pilot will not see you and the rotor blades at the back are also difficult to spot when they are in motion.
- Wear safety glasses and hearing protection.
- Avoid the rotor blades! Approach the helicopter crouching on level ground or uphill.
- Move away from the helicopter on a level road or downhill.
- Make sure the equipment you carry does not protrude above head height.
This is especially important in the vicinity of heavier helicopters, where the rotor blades can sink far down at low speeds due to their weight.
- Always secure anything that you suspect could be blown away before approaching the helicopter. Also consider any equipment and supplies in the landing area that could be blown away. Remove or secure the equipment/supplies.

In connection with water bombing, there is a risk of personnel being knocked to the ground, hit by stones that have been carried in the water load or branches that are pulled down by the water load. However, this risk only exists directly under the water load. There is therefore no reason to evacuate the fire area during extinguishing by helicopter, but instead ensure that the water bombing is carried out in cooperation with the emergency services personnel on the ground and keep an eye out for helicopters in the vicinity.

Also be prepared to temporarily step aside during a fall.

One of the differences between helicopters and airplanes in water bombing is that helicopters typically drop water with greater precision compared to airplanes.

This makes helicopters particularly useful for water bombing in close cooperation with ground personnel, including for safety reasons.

Also consider the safety of the waterbombing resources. Warn flying crews of hazards in the form of power lines, UAVs (Unmanned Aerial Vehicles) or drones in the air, etc.

11.13 Unmanned aerial vehicles – drones

Drone is an unmanned aerial vehicle in everyday language. The word drone is used by some organizations to describe third-party unmanned aerial vehicles; the Armed Forces do so in the event of antagonistic threats and the police in the event of suspected crime.



Figure 21. Example of an unmanned aerial system (UAS). Photo: Stefan Haggö.

Rescue services and police have chosen the name UAV (Unmanned Aerial Vehicle) to describe the aircraft itself and UAS (Unmanned Aerial System) to describe both aircraft, ground segment, ground control station and data link.

UAS offers great potential for expanded capabilities for the work of the emergency services. When used in municipal emergency services, UAS can – if used correctly – contribute valuable information. With the help of sensors such as cameras, infrared cameras, etc., fires and people can be located more quickly.

However, using UAVs today is not entirely without problems, as their use is subject to several different regulations. MSB has therefore produced guidance to support the implementation of unmanned aerial vehicles in municipal emergency services.¹²

With appropriate use of sensor data from a UAV, sensor data can contribute with valuable information, which can both facilitate the targeting and coordination of rescue efforts. In addition, these can contribute to more efficient and safer operations before, during and after a forest fire.

The benefit of UAS is considered to be greatest in connection with forest fires where it can be difficult to create an overview of the damage area from the ground.

¹² www.msb.se/sv/publikationer/obemannade-luftfartyg-i-kommunal-raddningstjanst-vagledning-2.0.

11.13.1 Airspace

In an airspace there may be several actors/operators and then there is a need for airspace coordination. This is to ensure that no incidents occur in the airspace, as there may be elements of both manned and unmanned flight in the same airspace.

Before a flight is carried out, the remote pilot must establish a flight and safety area, the extent of which is within the intended terrain and surroundings, and which must include the distance to people, animals and property. The main rule is that a flight in uncontrolled airspace must take place at a height lower than 120 meters above the ground or water. For flights in traffic information zones and traffic information areas, special provisions for radio communication apply. When flying in controlled airspace, special permission must be obtained from the relevant air traffic control unit for the airspace in question, so-called clearance. More about clearance, geo-zones and restricted areas can be found on the Swedish Transport Agency's information page on flying drones in airspace.¹³

Establishing a geo-zone creates opportunities to exclude unauthorized people and lock in authorized people.

When the flight area is secured, a secure connection must be established between the UAS function/commander and the emergency management/rescue management.

In the case of multiple flying actors/operators, an established airspace coordination can be established. Actors who can currently assist with this are the Armed Forces, the Coast Guard and the Police Aviation.

Due to the increased use of UAS in operations, the police have as a routine to always contact the operation command on the ground before entering the zone in which the operation is taking place. This is to ensure and minimize the risk of collision and considering unmanned aerial vehicles today have sensors/data that can be of paramount importance to the operation.

A UAS that belongs to the operation can be used advantageously in combination with other flying resources when an area is closed (temporary restricted area).

However, a need such as airspace coordination is a prerequisite for unmanned flight to be coordinated with manned flight. A secured connection between the flying resources is important. If a UAV is to fly at the same time as other flying resources, an airspace separation must be made laterally, and a direct connection must be ensured between the pilot and the remote pilot to avoid an incident/collision in the airspace.

11.13.2 Effects of forest fire

A UAS has several areas of operation, where an initial task may be to determine the extent and course of the accident. An early situational picture can both locate and guide incoming resources with route selection and size of the fire. In the initial risk assessment, a UAV can be used to locate power lines, businesses and homes that may be in immediate danger. If the aircraft system is equipped with an acoustic warning, it can also be used to signal danger to vulnerable people or inform that help is on the way.

13. www.transportstyrelsen.se/sv/luftfart/Luftfartyg-och-luftvardighet/dronare/flyga-dronare-i-lufttrummet/.

During a forest fire, a UAV can be used to check the fire ridge and flanks. An IR camera on a UAV is an excellent tool for identifying hotspots. Positioning marked on a map, or the area/coordinates visualized in another way, can be a support for then sending ground personnel to extinguish. A UAV can also, while operating in the airspace, film/

Photograph the burned area for documentation and feedback, where this material can later be used for learning purposes.

Another task is to monitor burned areas in real time and/or monitor the safety of deployed ground personnel.

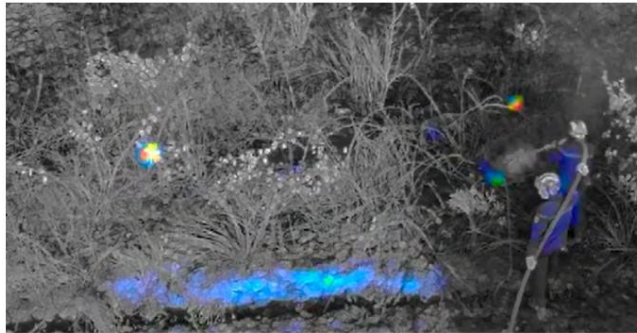


Figure 22. Example of detection via IR. Photo: IEDO14.

The emergency management/rescue management can decide to provide surveillance in different sectors. in the form of recurring overflights decided by the operational command. The remote pilot conducting the flight can detect hotspots in real time, and an optimization of the use of ground personnel can be achieved at the same time with this method.

11.13.3 Risks

Remote pilots must be aware of the strong convective winds that can be generated by the fire, as well as smoke that can make flight impossible by visual observation. Temperature fluctuations can also damage the drone's components.

A continuous risk assessment must be carried out at all times, as weather conditions can change quickly. Obstacles in the air, such as power lines but also trees, must be taken into account. A continuous analysis of obstacles must be carried out during forest fires, as the smoke can quickly turn. High-altitude vehicles can be used as a place for the remote pilot in order to detect obstacles and have better control during flight.

11.13.4 Sensors

There is usually a visual sensor (camera) on all flight systems, but many are also equipped with an IR sensor. With an IR sensor, an extended detection capability can be provided, and this capability is particularly good when the ambient temperature differs. If the remote pilots have night vision authorization, this can be used to great advantage at night. Smoldering fires in the ground and forest are identified quickly when the ambient temperature is lower, which can provide a better basis for the collected information, which can assist in the design of tactics the following day. There is often better space to fly in the airspace at night, when the crew is on the ground.

14. www.iedo-drone.org/archives/4799.

11.13.5 Tactics and management

In the case of established airspace coordination, the rescue command can advantageously establish a function called "Air Operations Leader". This person/function can act as a coordinator for the actor conducting airspace coordination and have the function of being responsible for safety between air and ground personnel.

It is also important that there is an established connection between the flying resources and whether there is a UAV in the area that belongs to the operation.

With a UAV in the air, monitoring and direct control of ground personnel can also take place in real time, as well as reading out the effect of measures or sources of error. In the case of long-drawn-out events, it may be good to set up a function with image interpretation/analysis. This function can act as a link between the remote pilot and the emergency management/rescue management.

If technical real-time transmission of images or film is available, support with analysis can be provided remotely, regarding the extent of the fire, the course of the fire, its spread, etc. This can be of great importance in creating an initial advantage. The technology for real-time sharing of video is particularly important for emergency management/rescue management on site. Current technology allows authorized personnel to see the image on their smartphone or mirrored on a screen at the control point. The goal is to provide situational awareness simultaneously in real time, which thereby provides a basis for the tactical plan, so that resources are moved to the best locations. This collected information can also serve as a basis for contingency planning with the aim of stopping the fire, slowing the fire or directing the fire to areas where it can cause less damage.

In addition to the technical part of sharing images, there are also challenges with regard to the legal aspects. For example, a distribution permit from the National Land Survey of Sweden is required in the acute phase to be able to share information outside their own organization.

11.13.6 GIS and map

Another use for a UAS and its sensors is to collect information for further processing and analysis. One such example is to refine the information by creating geodata that can then be used together with other information such as external GIS data. Taken together, it can be provided as qualitative support to decision makers and information for the situational picture.

An example could be mapping the spread of forest fires or floods. ning and thereby measure size and, for example, fire intensity or depth using image analysis. If the remote pilots have the ability to carry out a mapping, this can be carried out at an early stage. Many flight systems support so-called mapping/mapping. Mapping is carried out within the intended area, where a UAV flies and takes aerial images at different time intervals. The images are saved on memory cards which are then transferred to software to match the images together to an orthophoto also called an orthomosaic. The accuracy depends on the choice of sensor and the altitude at the time of the flight. Please ask the client about the accuracy (cm/pixels) before the flight is carried out.

Once the software has created an orthomosaic, it can be added as a layer on the background map, and/or printed as support for the response team.

Another way to create geodata is to send the GPS information from the UAS to a mapping application for digitization. By flying around an object, this information can be displayed as a point, line or area in the mapping application.

This information can be displayed on a background map in real time or saved in the map application. Another way is to use an application that can save flight metadata such as camera angles, heading and GPS coordinates.

The footage from the flight and the metadata are linked together in a GIS program so that the footage becomes geodata. The footage can then be used as a basis for generating points, lines and areas in a GIS program.

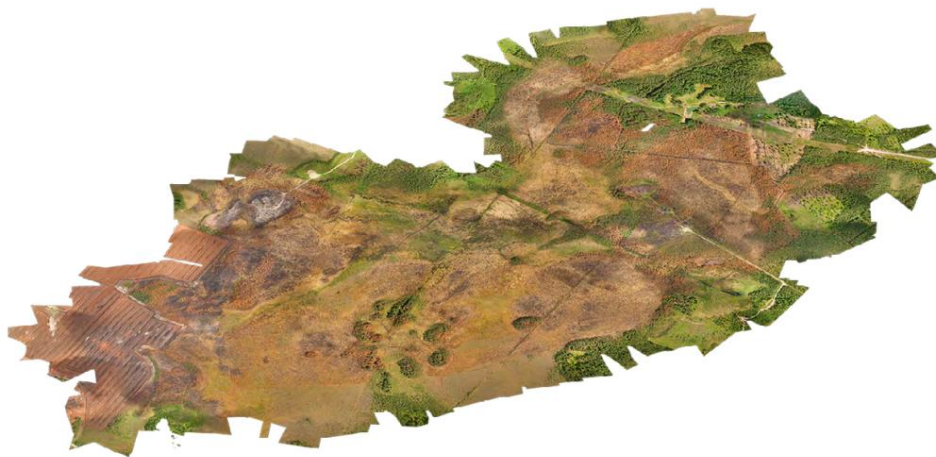


Figure 23. Orthophoto, taken one month after the forest fire, showing the burned areas at Flymossen in 2017. Photo: Henrik Lundqvist, Jönköping County Administrative Board.

The advantage is that you get an updated and more accurate map, where event-specific information is available for the fire, the surroundings and its spread. The information collected can later also be used as continued support for the operation, as well as documentation for feedback on experience and investigation of the progress and spread of the fire.

Data collected can also support 3D models or other calculations that can then be used for visualization.

If there is access to GIS support internally, use this resource early.

In the case of more complex accidents, the FSOL-GIS reinforcement resource can be activated via MSB's TiB. MSB's TiB activates and prioritizes/dimensions the resource, which in turn activates Lantmäteriet's Geocell via Lantmäteriet's TiB. FSOL-GIS can, for example, support by optimizing the use of geographic information for the situational picture, assist in the work with analyses and forecasts, visualize resources, create conditions for sharing geographic information and produce maps.



**Forest fire
near
buildings**

12. Forest fire near buildings

The first thing a fire officer does upon arrival at a forest fire that threatens buildings is to orient themselves and make an initial assessment of what direction the operation should take, taking into account the prevailing circumstances and available resources.

The same factors are considered as in a regular forest fire, but special risks that may exist in the building are also investigated, and the smoke from the fire can be pushed against the building and pose a risk to the residents. Is there perhaps a storage room or a workshop with gas cylinders or fuel containers? Where are the electrical lines?

The road system is also important to consider. Are there escape routes in case of fire? would be overwhelming? Is it possible to meet along the roads or are some roads basically one-way? It may be questionable to send vehicles to protect individual houses if there is only one way in. Vehicles must not end up in dead ends.

The commander must also consider how a possible evacuation will be carried out. A possible evacuation is easier to carry out during the day than at night. Furthermore, orders can be given to begin work on preparing an evacuation to save time once the evacuation is initiated. Residents can then be asked to prepare for a possible evacuation. Once the evacuation has begun, the chance of a successful evacuation is greater. In areas with a high risk of forest fires, the rescue service should have communicated the risk picture in advance and what measures the population can take in advance, to increase the speed of an evacuation.

Intervention planning in built-up areas is recommended, as the time frames during an operation in a building and associated evacuation may be limited.

The rescue leader must consider that the building may be a farm, for example. The operation may then be lengthy in terms of time and that additional resources may be required to assist with any evacuation of animals.

Consider positioning firefighting resources among the buildings in advance and early. Once the fire approaches the building, there may be rapid evacuations.

There is often a major shortage of resources in the initial stages of an operation, when sufficient resources have not yet been brought forward. Airborne resources, helicopters, are excellent in this situation, as they can be on site quickly and carry out extinguishing operations that delay the fire and create time for ground forces to carry out the hose laying. One piece of advice for the rescue leader in the event of a forest fire that threatens buildings is to have a helicopter circling over the fire throughout the operation. This will quickly identify where aerial fires and critical sections are located.

Once the fire commander has made his orientation and assessment at the scene, the commander must choose a defensive or offensive tactic when protecting buildings and fighting the fire (see more about this below).

12.1 Offensive tactics

The offensive tactic is used when resources and time are sufficient. The rescue service actively fights the forest fire before it reaches the settlement.

The emergency services personnel arrange containment lines and the idea is that the fire should never be able to come close to the building.

In addition to the resources used for active firefighting, the fire department should allocate fire engines and place them strategically among the buildings. The task is then primarily to combat aerial fires that may threaten the building, and to be ready to protect the building using defensive tactics if the fire's progress is not prevented.

The emergency services can of course also combine both tactics. Resources are then allocated to establish fire roads and containment lines from a distance, while fire engines are allocated to use defensive tactics to protect the buildings closest to the fire.

12.2 Defensive tactics

Defensive tactics are used when resources are scarce and time is short. This may involve the fire department distributing fire trucks and tankers to the threatened houses. Hoses are pulled from the trucks and a forest fire road is created around the houses.

The rescue service personnel wait for the forest fire to protect the building as the forest fire approaches. In the SiTaC system (see section 13.3), this hose system is called the "protection line".

Be mobile. When laying out hoses, do not automatically fill the hose system with water, but leave the system dry in case resources need to be quickly redeployed.

Three radiant tubes per house are usually sufficient. Two of the radiant tubes protect the house against the radiant heat and fights the approaching fire, the third jet pipe is used to wet the vegetation around the house and to fight the approaching fire. The firefighters, who are supposed to protect the house itself against the approaching forest fire, should wait to wet down, for example, the facade until there is less than half an hour left before the forest fire reaches it. It is better to save the water for actively fighting the fire itself. The jet pipes may also need to be used to protect firefighting vehicles.

Use extinguishing resources to combat any aircraft fires and fires in lighter fuels. This is to reduce the risk of ignition of the higher fuel layers which in turn can ignite buildings.

In the case of fires involving compact fuels, the rescue leader should allocate vehicles as a tactical reserve that can be quickly deployed to critical areas. This is motivated by the fact that water consumption increases in such fires.

Mobile sprinkler systems can be used to advantage when protecting buildings.

12.3 Measures to reduce the risk of ignition

Building protection agencies, as well as property owners, can increase the chances of saving the building through a number of measures. One of the first measures for such agencies is to reduce the amount of flammable material around the buildings, such as firewood, garden furniture, shrubs and trees, by removing it, wetting it down or covering it with tarpaulins. Also consider any accumulations of flammable material in gutters and on or under wooden decks.

It may be important to fell standing trees in order to reduce the flame height in the event of a forest fire, or to remove the lower branches to reduce the risk of fires in the higher fuel layers.

All doors, windows and ventilation should also be closed. Combustible materials inside the houses that are placed right next to the windows should be moved away from the windows. Also, cover the windows if possible, but make sure that it is easy to temporarily remove the covering to check and detect any ignition of the interior of the house.

Remove gas cylinders, such as gas grills, and containers with flammable liquids.



**Situational
picture,**

documentation and map

13. Situation, documentation and map material

13.1 Status monitoring

The need for situation monitoring is probably obvious to most people. Unfortunately, however, it is common to find improvised/temporary solutions and not be prepared when it turns out that staff work will suddenly have to be carried out in temporary premises, such as a conference facility, a school, etc.

However, the need for situation monitoring is no greater in the case of forest fires than in other major events. But because forest fires tend to occur far from fixed control points and also last for a long time, temporary solutions are sometimes required when the infrastructure is not in place. These temporary solutions are difficult to break because they require energy and steal time, which you do not consider to have in the intensive phase. When the situation has calmed down, you usually do not think it is worthwhile to change the system.

However, it is important when fighting forest fires that the same basis and situational picture are used and that, for example, different types of map material are not used.

The choice of map material should therefore be well thought out and preferably planned in advance.

Determine early in the operation which coordinate system will be used for positional information and inform all cooperating parties, helicopter crews, etc. about which coordinate system is being used.

In the event of a forest fire that has been going on for a long time, it may be a good idea to create map material with a separate grid system to make the extinguishing effort more efficient and minimize misunderstandings. It is important that all those involved are informed and use the same grid system and underlying map material. The grid system itself involves dividing the fire area into squares where, for example, letters are used for one axis and numbers for the other axis. The squares are then given individual designations – for example square D3 – which makes it easier, for example, when water bombing from a helicopter.

The grid system can be made in various ways, from ruler and pen directly on a map that is copied, to being done digitally via GIS software. Another simple way is to, for example, have a ready-made PowerPoint template with a grid and then insert the map image behind the grid. The grid does not need to have any positioning reference like RT 90 or WGS 84, but the main purpose is to function as a damage site map for communication for the actors involved. This map can of course be combined with plotting SiTaC symbols. In addition to the pure fighting phase, a grid system can also be used with advantage during the later phases when locating smoldering fires and securing the outer edges of the fire.

After the forest fire in Västmanland, the rescue services put in an enormous amount of work. on compiling and reconstructing the course of events and decisions, etc., because the information was scattered in paper notes, Google documents, etc.

One should therefore use a situation monitoring system that is set up and prepared in advance.

13.2 GIS

GIS (geographic information system) and geodata can be a powerful tool to use in a forest fire. In order to utilize the power of GIS and geographic analysis in an event, it is of great importance that the location of the event is known. The minimum should be a coordinate in the center of the event, but the best opportunities are given if the entire area of the fire is known.

Using a position for the fire, it is possible to perform a variety of GIS analyses and forecasts, for example to see which critical objects are in the vicinity of the fire and to be able to create fire spread maps where, using weather forecasts and vegetation maps, one can make a forecast of how quickly the fire can spread in different directions.

13.2.1 Copernicus

Copernicus is an EU programme that develops European information services based on Earth observation data from satellites and in situ data from sensors on the ground, in the air and at sea.

Copernicus is a user-driven initiative, primarily public authorities. The information services provided by Copernicus are free and available to users.

Copernicus provides information services in various areas, of which the Copernicus Emergency Management Service (EMS Mapping) covers events such as natural disasters, man-made events and humanitarian crises.

Sweden has made a number of Copernicus activations over the years. These have involved storms, forest fires and floods.

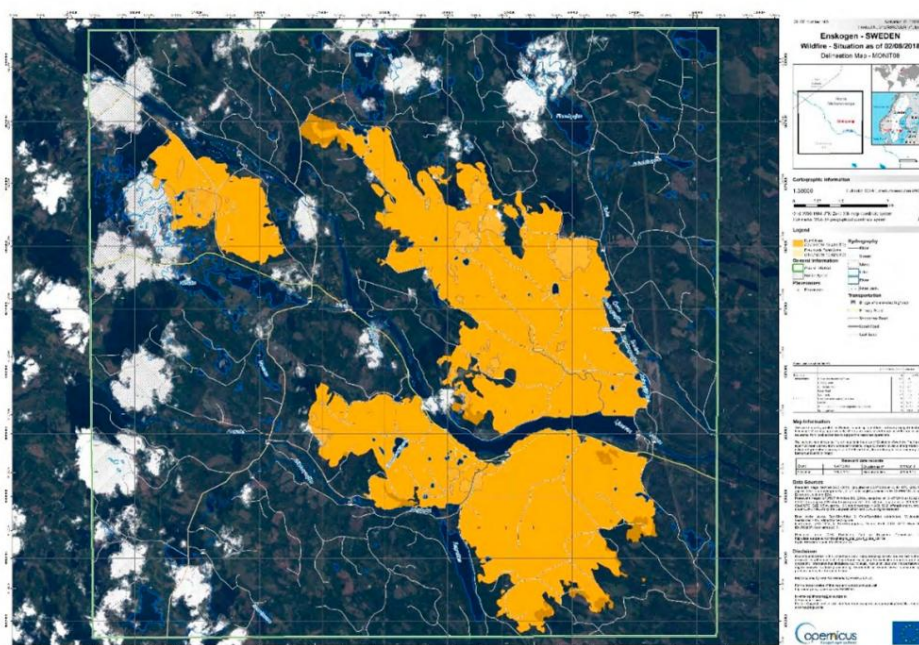


Figure 24. This is one of the maps produced during the forest fires in the summer of 2018, showing the spread of fire at a certain point in time.

Authorities, rescue services and municipalities can activate the Copernicus EMS service in the event of an incident by contacting the MSB's TiB. The MSB, which is the national contact point for the service, then forwards the activation to the ERCC (Emergency Relief Coordination Centre) in Brussels.

13.2.2 EFFIS

Within the Copernicus EMS (Emergency Management Service), a service has been developed for forest fire prevention. The service is called EFFIS (European Forest Fire Information System) and its purpose is to forecast fire risk in Europe and to map forest fires at an early stage. The website includes a so-called hotspot function where two different methods are applied (MODIS and VIIRS).

On the website (see link below) there is an application with a daily updated fire risk map across Europe with a time horizon of up to 9 days. FWI values and associated sub-indices can also be visualized on a map via the application.

EFFIS – Fire risk map

effis.jrc.ec.europa.eu/apps/effis_current_situation

In addition, the service provides statistics on fire history (distribution in space and time). EFFIS also offers support to those working on forest fire protection in the EU Member States and provides the European Commission and the European Parliament with up-to-date and reliable information on forest fires in Europe.

13.3 SiTaC

SiTaC is a system of map symbols to illustrate situational images and problems, but also tactical measures. When it comes to tactical measures, the system distinguishes between planned measures and ongoing measures.

The same applies to resources, i.e. the system distinguishes between planned resources and resources on site.

The system also contains symbols to describe the extent of the forest fire, its spread and factors that affect the behavior of the forest fire. The system also has the advantage that arriving resources can quickly get a picture of the situation, noted damage scene factors, objectives of the operation and planned and ongoing tactical measures.

The SiTaC symbols can be downloaded in a few different formats on the following website under the "Symbols" tab: krisberedskapskartor.msb.se.

See Appendix 4 for a more in-depth description of the system and the appearance of the map symbols.

13.4 MSB RIB Lupp

Lupp is a program for managing and monitoring rescue operations. Lupp is primarily aimed at Swedish municipal rescue services, for use at various levels from staff in the command center to operational personnel at the scene of an incident. The primary purpose is to provide a tool for accurate documentation of the sequence of events before, during and after a rescue operation.

Lupp will provide decision-makers with accurate, relevant and reliable information, along with forecasts of possible future scenarios and their consequences, leading to better decisions and more efficient emergency service work.

The basic functions in Lupp include Diary, Activity Overview, Status Table and Status Table. Through the MSB RIB Map program, the ongoing activities, including stations and units, can be visualized on a map.

Access to the SiTaC system is also available via MSB RIB Karta.

In the Diary, notes, status reports or decisions can be entered. In the Operations Overview, order is kept on ongoing operations and which units are out on assignment or available. In the Status Table, a simplified overview of the status of all ongoing operations is provided. The Status Table shows the status of units per station and thus provides an overview of the resource availability.

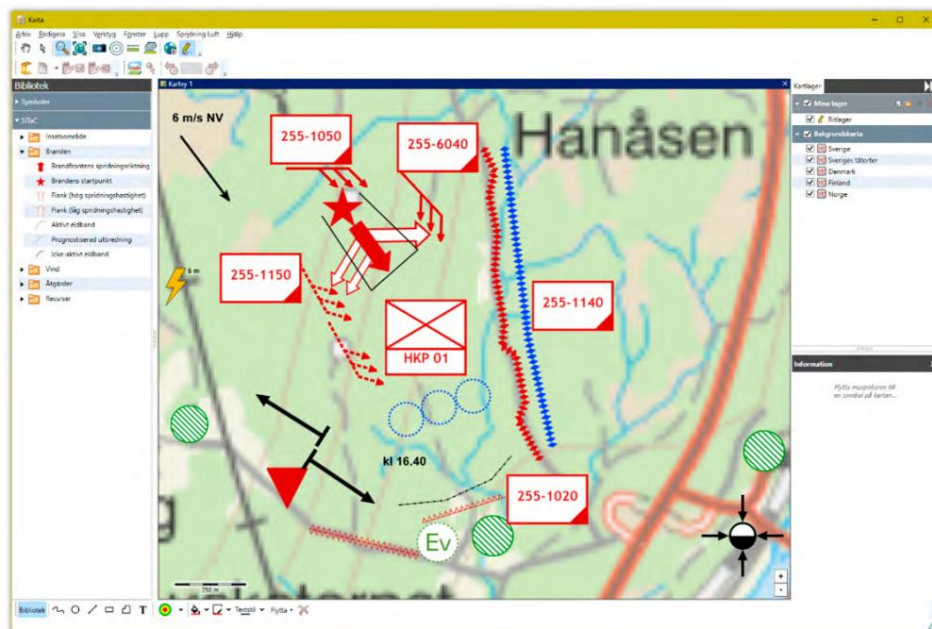


Figure 25. MSB RIB Map with SiTaC symbols.



Personal injury risks

14. Personal injury risks

14.1 Electrical hazards at power line street

Both electrical and non-electrical work near power lines involves various types of hazards. Those involved in firefighting in and near power lines must have basic knowledge of managing the electrical risks associated with firefighting work. Working with firefighting near a power line involves risks so that protective measures must be taken.

Previous regulations on electrical safety have been repealed ELSÄK-FS 2006:1 with reference to an employer applying the regulations for the work environment. Based on the work environment regulations, it is clear that through systematic work environment work, employers have the responsibility to detect situations where electrical hazards may arise and to take the necessary measures to manage these risks. A forest fire next to a power line street therefore requires that special protective measures are always carried out.

In previous regulations, certain distances were specified, which are reproduced in this text. However, it is important to consider electrical safety in every situation and to make a risk assessment and take protective measures. Measures in the event of a forest fire next to a power line should always be carried out according to the power line owner's procedures and instructions.

When fighting a fire in and near power lines, the line owner should always be contacted to be aware that work is underway and to be given the opportunity to take safety measures. However, there may be several different line owners within the extinguishing area. It is therefore advisable to establish contacts in advance between the rescue service, line owners in the area and the alarm center to clarify ownership, contact routes (telephone numbers) and procedures.

In the event of a forest fire next to a power line, it is important that the line is identified, so that the correct line is disconnected if disconnection is to be made. Larger lines can usually be identified using signs on the line poles where the line letter (information about the line) and the pole number are stated. This data should always be stated when contacting the line owner. At lower voltages, this marking is often missing and then localization must be done via GIS data, etc.

The rescue leader must always contact the line owner to determine whether the line should be disconnected (work without voltage) or not, considering the safety of the rescue service personnel during the extinguishing work. The line owner must receive supporting documents and a situation description from the rescue leader in order to decide on a possible immediate disconnection or shutdown of the so-called automatic reconnection for the line.

14.1.1 Safety distance

It is important that emergency personnel always maintain a safe distance when working near electrical installations. The distance should be determined from the nearest unprotected live part or conductor. Direct spraying of water onto live electrical lines should be avoided. Depending on the water quality, dangerous levels of current may be delivered to firefighting personnel and cause personal injury. Avoid working with water from a jet nozzle next to or near live electrical lines closer than 10 meters.

Special attention should be paid when transporting tools and vehicles on uneven ground next to or under a power line, as the load may swing out, change position, etc. as when bulky equipment is moved or lifted.

Distance based on previous rules for safety distances during non-electrical work (e.g. firefighting), special risk assessment is made in the individual case

Table 4. Lateral safety distances.

Voltage	Lateral safety distance
at low voltage (<1 kV)	at least 2 meters
at high voltage not exceeding 40 kV	at least 4 meters
at high voltage above 40 kV	at least 6 meters

Table 5. Safety distances in height.

Voltage	Safety distance in height
at low voltage (<1 kV)	at least 2 meters
at high voltage	at least 4 meters
at high voltage 400 kV	at least 4.5 meters

NOTE: The above safety distance does not apply in cases where the fire plume touches the line. In such cases, the safety distance is further increased and the safety distance also includes the pole and the fire plume.

The line owner should always be notified if fires occur near power lines, among other things to clarify whether a sufficient safety distance can be maintained.

If the distance cannot be maintained, the line needs to be de-energized or disconnected.

If the extinguishing work must be carried out within the safety distance for “Non-electrical work”, the extinguishing work is to be considered “Electrical work”. Such work must always be carried out under the supervision of a person appointed by the line owner/employer and competent for the work (the electrical work manager). The line owner ensures that the electrical work manager arrives on site and makes an assessment of the electrical risks that exist and how the work should be done.

in a safe manner. If the electrical supervisor determines that the line must be disconnected, extinguishing work may not begin within the safety distance until the electrical supervisor has notified the rescue leader that all measures have been taken to prevent electrical risk.

14.1.2 If the line cannot be disconnected

If the line cannot be disconnected, the work must be carried out with the electrician on site and taking into account the electrical risks that exist. The electrician must take the safety measures required to ensure that the extinguishing work can be carried out safely.

It is very important to maintain the specified distances to a live part, or the safety distance specified by the person responsible for electrical work. Rescue personnel must never enter the risk area with any part of their body, any tool or any other object during extinguishing work.

When extinguishing a fire near a power line or near an electrical installation, a dispersed water jet should be used, as a dispersed water jet has significantly lower conductivity than a closed jet. Contaminated water or salt water also significantly improves conductivity and thus increases the risk. Fire gases and water vapor from extinguishing work increase the risk of flashover. In the event of a flashover, a strong arc can occur that can cause severe burns.

14.1.3 Warning for automatic reconnection!

Some power lines have so-called automatic reconnection, which means that the line is automatically reconnected after a certain specified time. As a safety measure, the automatic reconnection should always be taken out of operation when extinguishing a forest fire next to a line. In addition, energizing a line can also occur during troubleshooting and sectioning.

It is important that emergency service personnel contact the line owner (contact the operations center or connection manager) to provide them with information to eliminate the risk of energizing through reconnection.

14.1.4 Fallen line

Keep in mind that a power line can sink several meters due to heat.

A downed wire should always be considered live until it is confirmed that the wire is disconnected and grounded.

NOTE: A downed power line can energize the ground over 10 meters. A person in that area can be killed by step voltage from the ground.

The risk of a downed power line coming into contact with a fence should also be taken into account. Tree felling near power lines should therefore not be carried out if there is a risk that trees could come into dangerous proximity to a live power line (step voltage).

Also consider the risk of a falling power pole as the pole structure may have been weakened.

14.1.5 Risk of explosion

Also consider the risk of explosion in electrical devices. This is greatest when extinguishing high-voltage devices that contain oil, such as a pole transformer.

14.2 Falling trees

The risk of rescue personnel being hit by a tree is high when extinguishing a forest fire. Trees that have been exposed to fire may have weakened root systems and fall more easily in strong winds or water bombing from the air. Even during simple helicopter overflights, there is a risk of falling trees.

There is also a risk of falling trees when securing the outer areas. Also be aware of the risk of falling trees when entering and exiting areas with tall and fire-affected trees.

Cut down trees that you think could pose a danger or mark them with chips and let someone else cut them down. But remember not to cut down all the trees after a forest fire, but only those that pose a danger. Dead trees after a forest fire are an important component for both flora and fauna.

14.3 Glödgropar

Glow pits are pits or cavities that contain large amounts of finely divided glowing material. They have high temperatures and pose a risk of burns if stepped on. Glow pits usually have no bearing capacity, and the foot can sink far down and be surrounded by glowing material. It can be difficult to see the glow pits as they often have a layer of white ash on the surface and it is not certain that you will see the embers that may be hidden further down in the ground. Natural cavities in the ground, anthills, peat and old coal beds (places where there used to be a coal mine) are examples of places where glow pits can develop.



Figure 26. Ember pit, Västmanland fire 2014. Photo: Mattias Strömgren.

14.4 High fire load

Areas with high fire load, for example logging areas with large timber stands, GROT stands (branches and tops collected after logging) or other accumulations of combustible biomass or logging areas, should be considered as risk objects and caution should be exercised there. If possible, prevent fire spread to these areas or objects.

14.5 Waterbombing

When water bombing, there is a risk that branches or undermined and fire-damaged trees may fall on rescue personnel due to falling water. Therefore, keep track of where helicopters or aircraft are and keep your distance from the point they are operating towards. Be prepared to step aside (helicopter) or move away from the immediate area (aircraft). Fire commanders must always keep track of where personnel are in relation to where the flying resources are water bombing. Helmets should be worn and high-visibility vests can be used to advantage when water bombing is taking place in the area.

14.6 Being surrounded by fire

During a forest fire, all emergency personnel may be inside the vegetation, which also constitutes the fuel. There is then a risk that emergency personnel may be surrounded by the forest fire, for example in the event of a windstorm.

The alternative is to immediately head to:

- The road where the fire truck is parked.
- A nearby stream or lake.
- An area that has already burned and is black.

Such options are called retreat sites, and it is important that all staff know in advance which retreat sites are available and where they are located.

In several countries, the abbreviation LACES (Lookouts, Anchor points, Communications, Escape routes and Safety zones) as a memory aid to avoid risky and unwanted situations for emergency personnel:

- Lookout refers to a scout who is placed in a strategic location to keep an eye on the fire and warn of flare-ups, for example. If possible, scouts in the air can also help, for example from a helicopter.
- Anchor point indicates the anchor point, i.e. to minimize the risk of the fire bypassing the rescue service personnel.
- Communications refers to the existence of functioning communication with neighboring units, scouts, flying resources, etc. that can warn of emerging dangers.
- Escape routes and Safety zones refer to places of retreat and Routes to these locations should be scouted, designated and known by personnel in advance.

14.7 Heat stress and dehydration

In the summer, temperatures are often high. Together with the heat from the forest fire, this can pose a danger to rescue personnel. During heavy exertion and heat exposure, large amounts of fluid can be sweated out.

If you don't drink enough fluids, you risk becoming dehydrated. Sweating is one of the body's defenses to keep your body temperature down.

When dehydrated, the body loses this ability and the body temperature rises.

If you continue to exert yourself and also expose yourself to heat, you risk eventually suffering from heatstroke.

Symptoms of heatstroke

- Body temperature above 40 degrees
- Nausea and vomiting
- Dizziness, irritability and confusion
- Fast pulse
- Red, dry skin

You need to get medical attention as soon as possible if you have heatstroke. It is important to get to a cool place immediately and get help to cool your body down, for example with a wet towel. In the hospital, you may need to get nutrients and salts directly into your blood.

To reduce the risk of heatstroke, you should drink more fluids. Pay attention if you urinate less than usual as this is a sign that you need to drink more fluids. It is also important to get enough energy and salts.

14.8 Appropriate clothing

To reduce the risk of heatstroke, emergency service personnel should avoid wearing thick, tight clothing when extinguishing a forest fire, unlike, for example, an apartment fire. A significantly better outfit is a thin forest fire overall with a cotton t-shirt underneath. Even better is a two-piece suit, which provides better ventilation.

A change of clothes is not to be despised. You stand there and get sweaty all day, and then in the evening when it gets cold it is an advantage to have something dry to put on.

On your feet, you should wear leather boots, preferably with high shafts and heat-resistant soles. Rubber boots are not suitable, as a forest fire often involves a lot of walking on hot days. Your feet feel better in leather boots than in rubber boots.

Make sure that, for example, sunscreen, mosquito repellent and chafing plasters are available. available in, for example, firefighting vehicles.

14.9 Appoint a security officer/coordinator

Consider appointing a security officer or security coordinator for a larger operation. The person in question should then be tasked with controlling the risks that exist and may occur and ensuring that safety work is coordinated. Risky elements can include, for example, protective burning or rocky terrain.

14.10 Impact of fire smoke

During the extinguishing of the forest fire, you inhale fire smoke, which contains carbon monoxide, tar particles, etc. Carbon monoxide is an odorless and toxic gas. When inhaling higher levels of carbon monoxide, you may experience headaches, dizziness and fatigue. Therefore, avoid exposing yourself to fire smoke as much as possible. Fire officers should remember that people exposed to constant fire smoke can become confused and get lost. Therefore, rotate personnel at regular intervals in areas with extensive smoke exposure.

Filter masks may be used. However, filter masks often remove irritants from the fire smoke, which otherwise has the positive effect of personnel moving away from the fire smoke early. However, since filter masks do not remove carbon monoxide and the irritants are removed, there is a risk that rescue personnel will stay in the area for too long and accumulate too much carbon monoxide. The only really adequate protection available is compressed air equipment, which is not always practical in a forest fire. Therefore, minimizing exposure can often be the best solution. Therefore, work with the wind at your back as much as possible.

14.11 Endurance

Forest firefighting is characterized by hard work over long periods of time. A number of measures can be taken to increase the endurance and efficiency of firefighting personnel:

- In addition to light personal equipment, personnel should also carry
Drinking water, preferably easily accessible in your belt.
- Personnel should be relieved at the jet tube to ensure continuity. When relieving in remote sectors, consider flying personnel out by helicopter to save valuable working time.
- Reinforcements should be brought in, for example by bus, train or plane (unless firefighting vehicles are also requested). Avoid emergency service personnel acting as drivers unnecessarily and arriving tired.
- Plan early for feeding emergency personnel. Initially, a simple food package may be sufficient, but prepared food should be provided early. Consider driving or flying food by helicopter to remote sectors.



Environmental considerations

15.Environmental considerations

When a fire escape is prepared, it should be laid out so that natural fire barriers can be utilized. This reduces the mechanical damage that occurs during the construction of the fire escape. Protective burning can also be used to reduce mechanical damage. However, it is not only nature and vegetation that can be damaged. Ancient remains can also be damaged by earthmoving machines, which occurred, for example, during the fire at Kräcklingbo on Gotland in 1992.

When using extinguishing water, injuries can also occur. Wildlife in a lake can be greatly affected if the water level is lowered because the shoreline is important for the lake's overall ecosystem. Therefore, you should monitor how much water is withdrawn and how it affects the water level. If you notice that the water level is changing significantly, you should change the watercourse. Also consider the risk of spreading crayfish plague. To prevent spread, disinfection of the material that comes into contact with the water is required.

Avoid using foam liquid admixture in the extinguishing water as this poses an environmental risk. Foam liquid admixture can have a negative environmental impact on watercourses even at low admixtures of a few per thousand, and also on vegetation at a few percent admixture. This is due to the toxicity of the foam liquid, which can kill fish, aquatic organisms and various plants. The toxicity in turn depends on the surfactant in the product. The degree of toxicity of different foam liquids can also vary greatly. The use of so-called long-acting retardants can also have a negative environmental impact.

When working to secure the outer areas, one should think before felling trees. Trees play a major role in wildlife, especially birdlife in the forest. Therefore, trees other than those that pose a clear risk to emergency personnel and people who will be in the forest in the future should not be felled.



Tactics for extinguishing forest fires

16. Tactics for extinguishing forest fires

This chapter has a summarizing role and is intended to show a common thread for the different components in the guidance but also to give the fire commander a pointer to where and when the different components are relevant during the initial stage of a fire.

The chapter largely covers the first three steps according to the so-called seven-step model¹⁵: reading the accident and making a risk assessment, creating a picture of possible measures and deciding on the objectives of the intervention and a tactical plan. The three steps have been described in the chapter from a forest fire perspective. To some extent, parts that are included in the other four steps are also mentioned.

16.1 Step 1. Read the accident and make a risk assessment

Reading the accident is largely about identifying what the need for help consists of. To be able to do that, you need to deal with the questions of what has happened, what is happening and what could happen.

When you receive an alarm about a forest fire, it is of great importance to get it right from the start. Therefore, take a moment when the alarm is raised to assess the information available. Someone needs to take on the role of reading the accident from the perspective of getting a good start on managing the operation. Especially in the case of forest fires, it is not uncommon to face challenges in finding the right place, how accessible it is, in which direction the front is spreading and whether there is anything at risk. This, together with finding a suitable breakpoint, can be things that commanders can think about before starting the emergency operation. The rescue services in the country handle this in different ways and the conditions are not always the same. The first personnel resources can, for example, be a first officer in their own vehicle who starts from home or an entire force that starts from a fire station with internal command.

In the case of forest fires, a lot of preparation and consideration can be done even before the fire occurs. During the summer, the fire risk forecast can be followed to, among other things, see trends and keep track of the current forecast. Feel free to include fire risk data and weather data from multiple sources to create the best possible situational picture.

In the event of high fire risk values, one should also consider which resources the alarm plan specifies and possibly make corrections. This may also involve prepositioning rescue forces in the event of high fire risk values. Based on the fire risk, one can also consider whether flying resources should be alerted immediately.

¹⁵Mattsson, M & Eriksson, L, 2010. *The Tactics Book – A handbook on how to implement effective interventions in the event of a fire in a building in a scenario-based manner*. Informationsbolaget.

During the approach, you continue to read the accident by taking part in the total information from SOS Alarm or equivalent. It is also possible that you will see the smoke from the scene from a few kilometers away. Color, thermals and turbulence can provide information about how intense and large the fire is. Weather parameters such as wind speed, wind direction, relative humidity and temperature should also be checked during the approach. Already on the way out, you can get indications of whether it is a forest fire beyond normal in terms of extent, spread, etc.

It is beneficial to mentally prepare yourself for what needs to be handled when you arrive. If you as a fire officer identify your possible short-term resource needs as early as possible, in addition to what has been alerted, it is also good to communicate this back to SOS Alarm/control center/ as early as possible.

internal command or similar. Also keep in mind that some resources may take longer to arrive at the scene, for example flying resources, and they may therefore need to be requested early. Resources can always be called back later if there is no need.

Once on site, the work of reading the accident begins, now with hopefully a better opportunity to verify the facts. Reading the accident in forest fires is often very challenging and difficult because of the terrain and vegetation.

It is difficult to see the entire damage area. In the case of larger forest fires, it is therefore of extra great importance to get an overview or, even better, a bird's eye view early on. Strive to get into the air as early as possible, if a helicopter is available, to get the best possible picture of the situation. Other alternatives may be images from forest fire flights or if a UAV with a camera or IR sensor for image transmission is available as an early resource. Regardless of whether you get the opportunity to read the accident from the air or not, the purpose is to gather information about the various damage site factors that you need to include in your assessment for future decisions.

In all orientation and reconnaissance, it is necessary to have access to good map material in order to be able to transfer what you see to landmarks in the terrain. If you have experience using the SiTaC model, you can use it at an early stage to your advantage.

- Draw the area and try to identify what is most important initially.
- Identify critical factors.
- Prioritize life, property (infrastructure, buildings, etc.) and the environment.
Please seek help from someone who is locally oriented.
- Seek collaboration with landowners and forestry companies and use them as resources for an overview and for continued efforts.

Let the reconnaissance take its time. The initial reconnaissance is important; a few extra minutes can save hours in the subsequent work, and it is easy to get stressed, especially if you are new to helicopter flying. It is good if the person who is going to get into the helicopter has flown before, so take the opportunity for environmental training if possible (for example, following the forest fire fighting aircraft during their regular flights). Feel free to take an extra round if necessary to get the best overview possible. Use the map for plotting during the overflight and pay special attention to protected objects, possible boundary lines, access roads, water intakes, any sector divisions, etc.

A risk assessment should be made during the initial reconnaissance, where, for example, the behavior of the fire is assessed and risks in the immediate area such as power lines, etc. are noted. For further information on risks when extinguishing a forest fire, see Chapter 14 of the guidance on risks.

16.1.1 Injury site factors

In the event of a forest fire, there are many factors at the scene that can affect how you respond. The following section highlights the factors that are typically the most critical. These are the ones you need to pay extra attention to.

Weather

This damage location factor is the most important to take into account when dealing with forest fires. Extreme weather conditions can cause serious consequences for the operation with a fire course that is difficult to influence, while at the same time it can expose us to great risks if we are in the wrong place. If you work proactively with this factor in the rescue service, you will probably have good control over the weather in both the short and long term during the forest fire season, even before the forest fire has occurred. This creates the conditions for being prepared for how the fire can be expected to behave. To get the parameters within weather that affect the operation, please use the MSB's information system "Fire risk forest and land" which is provided with data from SMHI.

Wind is the weather factor that most affects the direction and speed of a fire. If you want to know in which direction the fire will spread, you use the wind direction. For example, if there is a southerly wind, the direction of spread of the fire front will be north. The speed of the wind will have the greatest influence on the speed of the fire front. See also Chapter 5 on assessing the spread of a forest fire for rules of thumb for calculating spread speeds.

When looking at temperature and humidity in relation to each other, in some rare cases, you can achieve what is called a "crossover". This is when the temperature crosses the value for the humidity. Usually these are cases where the air temperature goes above 30 °C and the relative humidity goes below 30%. It is then particularly common for forest fires to occur and this can result in fire progression that is more difficult to manage. For further information on how weather affects forest fire behavior, see the book Forest Fire Extinguishing.

Temperature, relative humidity, wind and precipitation are the parameters that affect which FWI values you will get, which in turn will directly affect how the fire will progress. Wind has a strong impact on the fire risk value, but wind can also be difficult to forecast. This is important to keep in mind because the actual wind speed can be higher than the wind speed on which the fire risk forecast is based. It is also important for fire officers to understand how FWI works and affects a forest fire.

Vegetation and terrain

Vegetation, and above all ground vegetation, is the fuel of the fire. The type of vegetation where the fire band is located or threatens to spread to is a factor of great importance, which provides different conditions for which measures are possible and which fire course can be expected. For example, multi-layered vegetation can create a course that an older pine forest cannot give rise to. If you can identify which vegetation types are in the area, you can also use this tactically when, for example, you are choosing a location for measures, such as a direct attack or indirect attack or choosing the type of boundary line. The choice of which off-road vehicles you can use is also affected by the vegetation type. For the fire behavior of the different vegetation types, see the book Forest Fire Fighting.

The same applies when loading accident scene factors about the terrain. The first thing to identify is how the terrain affects the availability of the resources you have. Are there roads and if so, what types of vehicles can drive on them? Roads, but also paths, lakes and watercourses can also be used as natural boundary lines.

Topography

The topography, the slope of the ground, can affect your decision. You need to include this in your calculation of how quickly the fire will spread, but also that the fire behavior may change. A negative topographic change, i.e. when the slope goes from increasing topographic height to decreasing height, can be used tactically to advantage. Topography and directions that are favorable from an extinguishing point of view are downhill and northward slope, while uphill and southward slope are the opposite. For the influence of topography on fire behavior, see also the book *Forest Fire Extinguishing*.

Natural changes in topography or vegetation type are described as It is a sensitive point. This point can of course be both good and bad tactically. If you want to mark that point on a map, there is a special symbol in the SiTaC system.

The fire

As previously stated, it is possible to get a first indication of the intensity of the fire by studying the fire plume or the smoke column. When reading damage-scene factors about the fire as a commander, it is useful if it is possible to identify where the starting point has been and mark it on the map with, for example, a SiTaC symbol. This point can then be used as a reference in further assessments, including when assessing the rate of spread. It is also good if you can read where the different parts of the fire are on the map. You should identify where the back, flanks, wings and front are.

In most cases, low-level running fire is the main type of fire. If high-level running fire or perhaps even peak fire is observed, this needs to be taken into account with regard to the rate of spread and what measures may be possible. For further information on fire progression, see Chapter 4 of this guidance.

Boundary lines

Boundary lines can be divided into two categories: natural existing boundary lines and created boundary lines. Natural existing boundary lines can be, for example, lakes, watercourses, marshlands, power lines, roads, paths, certain types of deciduous forest, etc. These can be reinforced if necessary. Natural conditions for creating boundary lines should be used as it can save both time and resources.

Limitation lines can be planned in several stages. When planning your limitation lines, you can start with an inner limitation line where you choose to make a first attempt to prevent the spread of the fire. When work on the inner limitation line is underway, contingency planning should be started in case the inner limitation line falls. An outer limitation line can be planned and if you have sufficient resources, work can also begin there. Ultimately, contingency planning should include a worst-case scenario, in which case the outer limit should also

the containment line would fall. A worst-case scenario, for example, could mean a last chance to prevent the fire from spreading before it reaches buildings or other important infrastructure.

The task force can provide valuable support by, among other things, identifying possible containment lines, and the force also has the opportunity to analyze and take into account to a greater extent the factors that affect the possibilities of establishing a containment line, such as resource availability, weather, speed of spread and time aspects.

For further information on containment lines, ground and aerial control, see Chapter 7 of this guidance.

People

When a fire occurs in a building, it is natural to ask whether any people are threatened or could be threatened by the incident. As long as this is not known, measures are usually focused on ensuring that this is not the case.

Nature is a common place for people to be and it is therefore important to consider this when reading the accident. If people are threatened, this must be dealt with first. Carry out any evacuations in collaboration with the police or ask them to prepare for the evacuation of threatened areas. Block off access roads into the area. For a planned or completed evacuation, there are special symbols in the SiTaC system.

In connection with evacuations, communication with the media and the public plays a central role. The key in this situation is to get out accurate information. Consider requesting reinforcements to handle the increased demand for communication.

Environment

The environment is a factor that can be very significant geographically and in the long term. As a commander, you need to make an early assessment of what is threatened in the immediate area. In the early stages, it could be a holiday home, a barn or other building. It could also be ancient monuments, biotopes or other things worth protecting in that area. A power line in the area could be a socially important function that is threatened, as could a motorway, a hospital or other critical dependencies.

Once you have dealt with the critical immediate area, it is important to also think in a longer term perspective.

Risks

From the earliest stage, it is important to identify actual and potential risks.

Identified risks can, for example, be a fire process that poses or may pose risks to personnel, terrain that poses a risk of falling accidents, etc. The risks you identify affect which of your identified possible measures (step 2) you ultimately choose to use in the tactical plan (step 3).

16.1.2 Development of events

A key factor in extinguishing a forest fire is to constantly work in different time scales, i.e. to try to have both a short and a longer time perspective on the development of events, since forest fires are usually long-drawn-out in time. For the development of events, it is important, for example, to weigh up factors that affect the behavior and course of the forest fire, such as weather, topography, fuel, the advance of extinguishing resources in the terrain and the planned arrival of flying resources. Here, for example, rules of thumb for assessing the likely spread and extent of the fire can serve as support (see further in Chapter 5 on assessing the spread of a forest fire).

16.1.3 Risk assessment

This step includes identifying the risk, assessing its likelihood and the potential consequences if it occurs. It also includes considerations related to managing the risk, such as containment, zoning, and setting the level of protection and restrictions for those who may be exposed to the risk.

Specific risks in the event of a forest fire may include, for example, power lines, fire smoke and heat effects. Risk-reducing measures may include, for example, ensuring that each sector has at least two possible ways out of the area or that all personnel are aware of the applicable retreat locations. For further information on risks and risk-reducing measures, see Chapter 14 on risks.

When assessing risk, take into account both current risks, impacting risks and risks that do not currently exist but that may arise if conditions change. The latter case could, for example, be a wind capsize or an increase in intensity during a transition to high-speed fire.

16.2 Step 2. Identify possible actions

16.2.1 Resource access

Landowner

Establish contact with landowners as early as possible in the operation. The landowner has good knowledge of the stand and terrain conditions and can indicate property and areas with special values that may be of importance for the planning of the operation. If there are many smaller private landowners in the fire area, coordination may be necessary. Coordination can then be done by the landowners themselves appointing a representative to represent them, a residual value manager, a forest owners' association, a forestry company or similar.

If a larger forestry company is the landowner, there are often contacts and resources available that can be of great importance for the implementation of the operation. This can be mapping material, harvesters, forwarders, construction machinery, personnel, own fire-fighting equipment and the like. A good resource when planning boundary lines are the planners who are attached to the forestry companies. They can be of great help when it comes to, for example, assessing the time required for a certain measure or marking planned fellings to create or reinforce existing boundary lines.

A good way to get started with landowner collaboration is to initiate landowner meetings early on where you can check on the work and plan ahead. The meetings should

be held regularly, preferably at a fixed time each day. Involving landowners in the operation early also facilitates the work after the rescue service has ended.

They are then already familiar with the work and have had time to plan for the continued work after the rescue services have finished.

For further information on securing the outer areas, completion of rescue operations, service etc., see chapter 10 of the guidance.

Forest fire specialists

Personnel with specialist knowledge in forest fire control are a good resource and can be of great help in decision-making. Burn leaders often have experience from both controlled nature conservation burns and forest fires and can assist with suggestions for measures and methods that can lead to a sustainable tactical plan. When establishing a protective burn, special consideration should be given to their experience. Burn leaders are found, for example, at county administrative boards, forestry companies and private contractors.

MSB's reinforcement resources

If rapid containment measures are needed in the early stages of fire spread or in the event of larger, resource-intensive forest fires, MSB's reinforcement resources are available as an alternative. The reinforcement resources include airborne support with helicopters and bucket planes, command and coordination capabilities, forest fire depots and high-capacity pumps.

Flying resources

Aerial firefighting may, at certain stages of a forest fire, be the only extinguishing measure that is currently possible depending on resource availability, smoke development, risks to ground personnel, inaccessible terrain and in critical situations.

In most cases, however, the flying resources complement the efforts made on the ground.

Helicopters generally require little preparation when it comes to landing sites etc. When extinguishing a fire by helicopter, ground personnel can usually remain in the respective sector. It is then an advantage if the ground personnel are equipped with high-visibility clothing.

Water-bombing aircraft require greater preparations, partly because they need larger watercourses and lakes to fetch water from. For each type of aircraft, there are different minimum requirements for the depth, length and width of the lake or watercourse in question. According to Chapter 6, Section 2 of the Swedish Maritime Safety Act, the rescue leader may, during a rescue operation, prepare himself and participating personnel for access to another person's property, cordon off or evacuate areas. However, cancellation of watercourses cannot take place before an ongoing rescue operation with the support of the Swedish Maritime Safety Act. Cancellation of watercourses must be documented according to Chapter 6, Section 6 of the Swedish Maritime Safety Act, i.e. a decision in writing stating when and by whom the decision was made, the reasons for the decision and what the decision concerns. Cancellation of public waterways should be avoided as much as possible, but if cancellation of watercourses that constitute public waterways takes place, this must be communicated to shipping in good time. This communication is most conveniently conveyed via the following contact channel:

Sweden Traffic: swedentraffic@sjofartsverket.se ; tel: 0771-63 06 85

In the event of cancellation of a public fairway, the Swedish Maritime Administration must also be informed promptly. via its TiB function which can then forward the information internally.

The Swedish Maritime Administration's TiB is easiest to reach via SOS Alarm.

The Swedish Maritime Administration also recommends that consultations take place, if possible, to determine a suitable location for water collection from a maritime perspective.

If necessary, the relevant airspace can be illuminated after a decision by the rescue leader. Applications for restrictions on aviation are sent to the Swedish Transport Agency. Remember that no-fly zones also apply to drones. For further information on waterbombing tactics, cooperation with ground resources, etc., see Chapter 11 on flying resources in the guidance.

Depot location

Designate a depot location where resources and equipment can be gathered. It should be a central location that is easily accessible and has plenty of space. It is advisable to appoint a logistics manager.

16.2.2 Possible measures

Creating and reinforcing boundary lines

A large operation requires a varied approach. When choosing a measure, the conditions of the location should be taken into account. When dividing an operation into sectors, the method(s) to be used to create boundary lines should be taken into account. Different methods can be combined. For a successful result and optimized utilization of resources, a varied approach is often required based on the prevailing conditions at the location such as terrain, weather, accessibility, road network, water supply, etc.

A boundary line can be built in several steps:

1. Designate an appropriate natural boundary line.
2. Widen by clearing or felling.
3. Transport the wood (combustible material) away from the boundary line.
Here, fire-damaged material is separated, which is placed in the burned area, and unburned material, which is placed in the unburned area.
4. Reinforce the containment line through water spraying, protective burning or excavation depending on resource availability and how the fire develops.

The last point in the above list is in some cases referred to as "blackline".

"Blackline" means securing the burned area with water spraying and the name refers to the fact that the gray (burned) turns black when water is sprayed. Blackline can be done around the entire fire area before handing it over for guarding. The method is done in several steps, where the first step is to secure 20 meters into the burned area. Blackline can be applied in hard-to-reach areas because it can be carried out with only ground personnel.

For further information on boundary lines and ground control, see chapter 7 of this guide.

16.3 Step 3. Decide on MMI and tactical plan

Just like in other types of rescue operations, the MMI, the goal of the operation, is one of the most important decisions you make as a commander. It may seem that the goal is obvious, but it often turns out during operation evaluation that the lack of a stated goal can cause hesitation and that there are different perceptions about why different actions are taken. MMI is therefore about what you want to achieve or what you do not want to happen.

To be able to achieve this goal effectively, you need to implement a tactical plan. The tactical plan describes how you want to achieve the goal.

Be sure to communicate both the MMI and tactical plan to the sectors in the forest.

16.3.1 MMI (Goal with effort)

What might a decision on MMI look like when it comes to forest fires? What do you want to achieve? An MMI should preferably handle the entire affected context.

It might be useful to try turning the concept around to instead describe what you don't want to happen because it often makes it easier to produce an MMI. Some examples might be that the forest fire should NOT spread to the eastern side of the road, that the community in the direction of the front's spread should not be affected, that personnel should not be injured, etc.

16.3.2 Tactical plan

To concretize a decided MMI, a tactical plan is developed. This describes HOW we will achieve MMI and is often expressed with the mnemonic rule IDA (Introduction... Then... Concluding).

A tactical plan can be designed differently depending on its scope and who it is intended for. A tactical plan can be made for the entire operation, and thus reflect the MMI for the entire operation. BUT a tactical plan can also be designed specifically for an individual sector and then becomes a little more detailed for their tasks so that it becomes easier for a sector commander to execute.

When developing the tactical plan, you should establish an inner and, if possible, outer boundary line.

Example of a tactical plan for the entire operation:

I: Create a situational picture, take stock of what needs we have in the short and long term...

D: Put out the fire...

A: Support and help those affected to get started with aftercare...

Example of a sector tactical plan:

"The goal of the operation is to prevent the fire from spreading to the opposite side of the road."

This should be achieved by initially delaying the spread of fire along the fire front and flanks using water-bombing resources, in order to create time for ground resources to advance in the terrain. Then, direct attacks should be initiated against the respective flank and wing while flying resources concentrate their attacks against the wings in order to compress the fire front. Finally, direct attacks should be carried out against the fire front; the flying resources suppress any flare-ups along the fire front."

16.3.3 Metodval

The tactical plan usually includes a number of different method choices to achieve the goal of the operation. Method choices may include, for example, the construction of a fire escape along a certain stretch, water bombing along a critical section, or the execution of protective burning.

Choose methods depending on the prevailing conditions. What resources are available in the short and long term? Which methods will be most effective in the specific case? Vary the methodology based on terrain and conditions. Feel free to create a toolbox that includes, for example, hose laying with jet pipes, sprinklers, blackline, water bombing resources, bog trampling with tracked vehicles, irrigation from the road with agricultural machinery, road construction, excavation and protective burning.

16.3.4 Security

Possible safety aspects could include, for example, alerting staff to the fact that the fire has caused some rushes and that flare-ups are frequent, and informing them of the appropriate retreat locations. Consider appointing a safety officer early on in

the event of a major forest fire who is responsible for safety. Make it a habit to inform staff entering the fire area about the risks and work environment when they check in.



Appendix 1: Form, request for support from MSB's reinforcement resource helicopter or aircraft



Myndigheten för
samhällsskydd
och beredskap

Checklist for requesting airborne support for forest firefighting, 2022

The request is made by the rescue leader or county administrative board directly to MSB. Anyone requesting support should contact MSB via the Civil Servant on standby (TiB) on telephone: 054 -150 150 and send the completed checklist below to: tib@msb.se

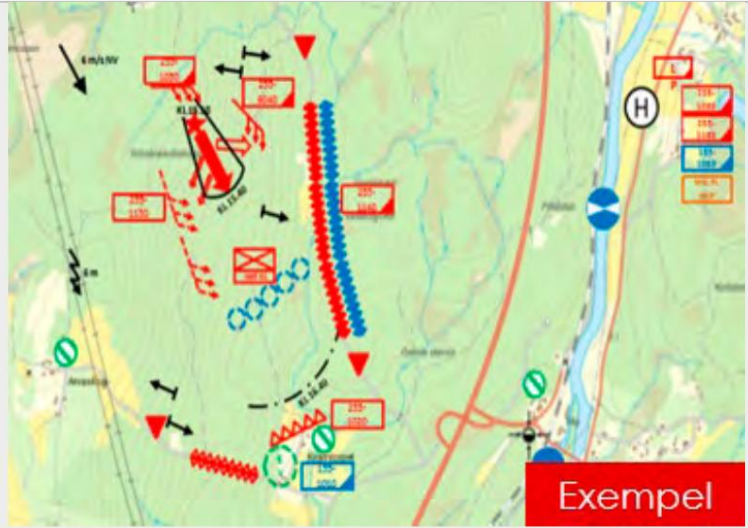
The MSB typically sends reinforcement resources immediately after a call between the requesting party and the MSB. However, the checklist must be promptly submitted to MSB as a receipt upon request.

Information on page 1(2) is mandatory.

Mandatory information:

Contact details for the competent decision-maker for the request:	Management system: Click here to enter text. Name: Click here to enter text. Phone: Click here to enter text. Rakel: Click here to enter text. Email: Click here to enter text.
Contact information for the person/role leading the response at the scene:	Name: Click here to enter text. Phone: Click here to enter text. Rakel: Click here to enter text. Email: Click here to enter text.
Location/town and date:	Place for insert: Click here to enter text. Municipality: Click here to enter text. Date and time: Click here to enter text.
Coordinates to the incident/incident site: NOTE: Use only one of the reference systems, e.g. WGS84 or SWEREF99 TM, specify which one.	Click here to enter text.
Risks for flying resources?	Drone ☐ Power line ☐ Radiomast ☐ Other flying resources participating in the operation ☐ Other ☐ Click here to enter text.

Non-mandatory information:

Estimated area:	Click here to enter text.
Risk picture:	Threat to life or health ÿ Power lines ÿ Electricity supply ÿ Radiomaster ÿ Building ÿ Forestry ÿ Other ÿ Click here to enter text.
Top fire?	Yes ÿ No ÿ
Aircraft fire?	Yes ÿ No ÿ
Other information to MSB:	Click here to enter text.
Map of the operation area/SiTAC map. Replace the example image on the right.	



Appendix 2:

Form, request for support from The Armed Forces



REQUEST FOR SUPPORT FROM THE DEFENSE FORCES

☐
ACCORDING TO THE SUPPORT REGULATION (2002:375) or FFS 2002:7 §3(Does **not** apply to rescue services)
☐
ACCORDING TO THE ACCIDENT PROTECTION ACT (2003:778)

The request for support concerns (briefly describe the capabilities/resources requested and the event):

Law/regulation:

Ordinance (2002:375) on the Armed Forces' support for civilian activities

The Swedish Armed Forces' regulations (FFS 2002:7) on the Swedish Armed Forces' support for civilian activities (§3)

Act (2003:778) on protection against accidents (LSO)

	a. To be completed by the authority/actor requesting support:	Signature
Date and time of request		
Authority/actor		
Petition decided by (position and name)		
Direct number to authority/staff/actor		
Make an assessment regarding the issue of situations that may arise where FM's personnel may use coercion or violence against individuals (SFS 2002:375 §7)		
Make an assessment regarding the question of FM's personnel may be used for tasks that may entail a significant risk of injury to the personnel (SFS 2002:375 §10)		

FÖRSVARSMAKTEN

Datum

20 -

Designation

Like 2 (3)

Other:

In the case of support from the Swedish Armed Forces (2002:375; FFS 2002:7 §3), the applicant is obliged to compensate the Swedish Armed Forces for damage that occurs to borrowed equipment during the loan period. The applicant is obliged to restore borrowed equipment in the same condition as when it was loaned at a location designated by the Swedish Armed Forces. A fee is charged by a government authority in an amount that at least corresponds to the additional costs of the Swedish Armed Forces. For others, a fee is charged in an amount corresponding to full cost coverage.

Under LSO (2003:778), no compensation is paid to the Armed Forces.

Time no.: b. Info/measures are filled in by the Armed Forces

The petition was submitted to VB Regional Staff North, Central, West, South

Regional staff decisions, FragO etc.

The petition was submitted to VB FM

Any comments from JURS:

INS Operations Manager's decision:

FÖRSVARSMAKTEN

Datum

20 -

Designation

Like 3 (3)

The form is sent by the Central Authority/equivalent digitally to the Officer on Duty, Swedish Armed Forces (VB FM) via email: vbfm@mil.se or via open fax 08-788 77 78 (HKV liaison center).

For information or questions, contact VB FM Phone 08-788 81 11.

The form is sent by the regional/local authority or actor digitally to the respective regional staff below:

Duty officer at Regional Staff North (VB RS N) within Jämtland, Norrbotten, Västerbotten and Västernorrland County:

Email address vb-mrn@mil.se, telephone 0921-34 91 00, fax 0921-34 84 45, fax 0921-34 93 45.

Duty officer at Regional Headquarters Central (VB RS M) in Dalarna, Gotland, Gävleborg, Stockholm, Sörmland, Uppsala and Västmanland counties: Email address mrm@mil.se, telephone 08- 58 45 40 42, fax 08-58 45 40 80, fax 08-58 45 40 90.

Duty officer at Regional Staff West (VB RS V) in Halland, Värmland, Västra Götaland and Örebro counties: Email address vb-mrv@mil.se, telephone 0500-46 61 00, fax 0500-46 61 01, fax 0500-46 61 02.

Duty officer at Regional Staff South in Blekinge, Jönköping, Kalmar, Kronoberg, Skåne and Östergötland counties: Email address vb-mrs@mil.se, telephone 046-36 88 00, fax 046-36 80 50, fax 046-613 76.

Always make sure the request has been received by calling and confirming!



Appendix 3: Application form, request for support from The Armed Forces



The person requesting support is required to tick the type of support intended.

REQUEST FOR SUPPORT FROM THE DEFENSE FORCES

☐

ACCORDING TO THE SUPPORT REGULATION (2002:375) or FFS 2002:7 §3

(Does **not** apply to rescue services)

☐

ACCORDING TO THE ACCIDENT PROTECTION ACT (2003:778)

The request for support concerns (briefly describe the capabilities/resources requested and the event):

Here you fill in the type of resource you need support with.
(e.g. munitions clearance resources, personnel, wheel loaders, etc.) and a brief description of the problem.

Law/regulation:

Ordinance (2002:375) on the Armed Forces' support for civilian activities

The Swedish Armed Forces' regulations (FFS 2002:7) on the Swedish Armed Forces' support for civilian activities (§3)

Act (2003:778) on protection against accidents (LSO)

	a. To be completed by the authority/actor requesting support:	Signature
Date and time of request	Fill in the date, time, which authority is requesting support, who has made the decision to request support and their phone number. DON'T FORGET TO SIGN!	
Authority/actor		
Petition decided by (position and name)		
Direct number to authority/staff/actor		
Make an assessment regarding the issue of situations that may arise where FM's personnel may use coercion or violence against individuals (SFS 2002:375 §7)	Make an assessment of whether situations may arise where FM's personnel may use coercion or violence against individuals. At least YES or NO must be filled in. DON'T FORGET TO SIGN!	
Make an assessment regarding the question of FM's personnel may be used for tasks that may entail a significant risk of injury to the personnel (SFS 2002:375 §10)	Conduct a risk assessment to see if situations may arise where FM's personnel could be harmed. At least YES or NO must be filled in. DON'T FORGET TO SIGN!	

FÖRSVARSMAKTEN

Datum
20 -

Designation

Like 2 (3)

Other:

In the case of support from the Swedish Armed Forces (2002:375; FFS 2002:7 §3), the applicant is obliged to compensate the Swedish Armed Forces for damage that occurs to borrowed equipment during the loan period. The applicant is obliged to restore borrowed equipment in the same condition as when it was loaned at a location designated by the Swedish Armed Forces. A fee is charged by a government authority in an amount that at least corresponds to the additional costs of the Swedish Armed Forces. For others, a fee is charged in an amount corresponding to full cost coverage.

Under LSO (2003:778), no compensation is paid to the Armed Forces.

Time no.: b. Info/measures are filled in by the Armed Forces

The petition was submitted to VB Regional Staff North, Central, West, South

Regional staff decisions, FragO etc.

TO BE FILLED IN BY MR S

The petition was submitted to VB FM

Any comments from JURS:

FILLED IN BY HKV

INS Operations Manager's decision:

FILLED IN BY HKV

FÖRSVARSMAKTEN

Datum

20 -

Designation

Like 3 (3)

The form is sent by the Central Authority/equivalent digitally to the Officer on Duty, Swedish Armed Forces (VB FM) via email: vbfm@mil.se or via open fax 08-788 77 78 (HKV liaison center).

For information or questions, contact VB FM Phone 08-788 81 11.

The form is sent by the regional/local authority or actor digitally to the respective regional staff below:

Duty officer at Regional Staff North (VB RS N) within Jämtland, Norrbotten, Västerbotten and Västernorrland County:

Email address vb-mrn@mil.se, telephone 0921-34 91 00, fax 0921-34 84 45, fax 0921-34 93 45.

Duty officer at Regional Headquarters Central (VB RS M) in Dalarna, Gotland, Gävleborg, Stockholm, Sörmland, Uppsala and Västmanland counties: Email address mrm@mil.se, telephone 08-58 45 40 42, fax 08-58 45 40 80, fax 08-58 45 40 90.

Duty officer at Regional Staff West (VB RS V) in Halland, Värmland, Västra Götaland and Örebro counties: Email address vb-mrv@mil.se, telephone 0500-46 61 00, fax 0500-46 61 01, fax 0500-46 61 02.

Duty officer at Regional Staff South in Blekinge, Jönköping, Kalmar, Kronoberg, Skåne and Östergötland counties: Email address vb-mrs@mil.se, telephone 046-36 88 00, fax 046-36 80 50, fax 046-613 76.

Always make sure the request has been received by calling and confirming!

Scan or fax the form to one of these addresses, call VB MR S to ensure that the form has arrived.

Appendix 4: SiTaC

UB-RTJ
Rickard Hansen

rickard.hansen@msb.se

SiTaC - map symbols for forest fire extinguishing

Introduction

In 2015, a number of representatives from MSB, municipal rescue services, county administrative boards, the Swedish Forest Agency, SMHI and MSB participated in a forest fire training course in Italy. They brought back with them, among other things, the SiTaC system.

SiTaC is a system with map symbols to illustrate the situation, problems and also tactical measures. When it comes to tactical measures, the system distinguishes between planned measures and ongoing measures. The same applies to resources, i.e. the system distinguishes between resources on site and planned resources. The system also contains symbols to describe the extent of the forest fire, its spread and factors that affect the behavior of the forest fire. The SiTaC system actually covers all types of events, but one limitation, for Sweden at the moment, is to only include the symbols used in forest fire extinguishing.

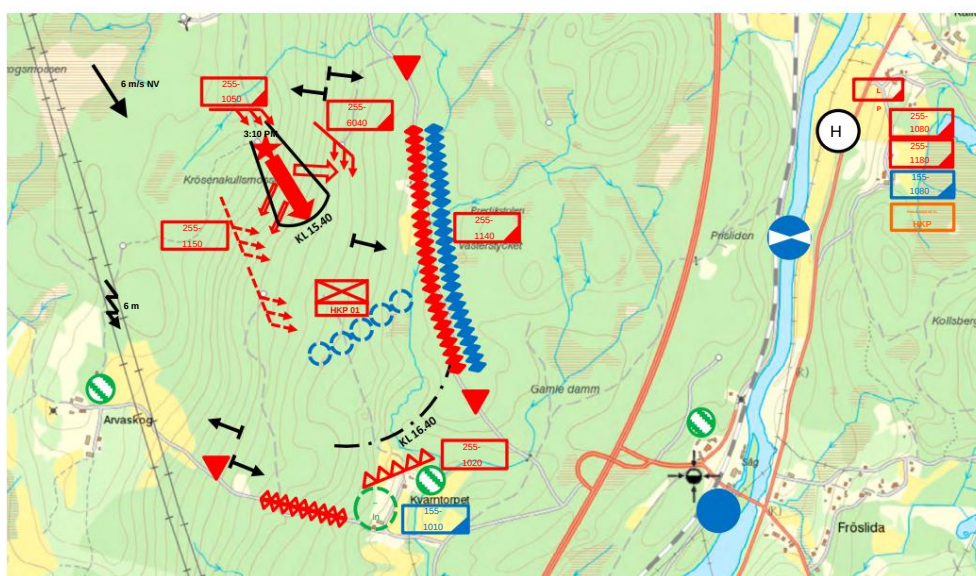


Figure 1. Example of map image with SiTaC symbols.

SiTaC stands for Situation Tactique in France and Situazione Tattica Complessa in Italy. The system has been developed and used by Mediterranean countries for several years.

Given the high risk of forest fires in these countries, the map symbols for forest fires are particularly well developed and thought out.

Since forest fire extinguishing is primarily a field activity with field conditions, all graphic symbols used have been developed to be easy to understand and also work in black and white presentation.

The SiTaC system with map symbols also includes an analysis component, more specifically a catalogue of rules of thumb to create an image of how the forest fire spreads over time and along the terrain. These rules of thumb can be found in the MSB's forest fire extinguishing guidance.

What are the benefits of SiTaC for Sweden? With the SiTaC system, the use of reinforcing flying resources from southern Europe will be made more efficient as the flying crews usually request a situational picture illustrated with the SiTaC symbols.

An implementation of the SiTaC symbols in Sweden also has the advantage that reinforcing resources from another part of the county, the rest of the country, from another authority or organization can quickly get a picture of the situation, taken into account damage site factors, objectives of the operation and planned and ongoing tactical measures. In this way, understanding of the firefighting operation itself increases and the safety of personnel on the ground and in the air increases, where everyone involved in the operation can see in a simple and transparent way that all resources are deployed.

The SiTaC system with its map symbols would also raise awareness among Swedish rescue services regarding influencing factors, possible tactical measures, etc. when extinguishing forest fires.

This memo has been prepared by a working group consisting of representatives from MSB, Halmstad Rescue Service, Sävsjö Rescue Service, Eastern Blekinge Rescue Service and the County Administrative Board in Jönköping County.

Usage and description

The symbols can be applied to a digital map, drawn directly on a paper map or by hand on blank paper/whiteboard.

The symbols are different in nature, with some symbols simply being enlarged or reduced while other symbols have different designs and sizes depending on the design and extent of the insert and are more characterized by freehand drawing.

When using the SiTaC system in the initial stages of a forest fire, the system should be seen as a rough tool where, among other things, the placement of the various symbols on the map material does not necessarily need to be precise.

Situation

Within this area of application, symbols are used to describe:

- Deployed and planned resources: type of resource and the resource's geographical location. The resources are indicated as boxes with a possible call sign or with abbreviation/plain text (other organization or flying unit).
- Management site.
- Breakpoint.
- Description of the fire area and nearby areas.
 - o Suitable water intakes, their locations and a description of what type of extinguishing resource the water intake in question is suitable for or intended for: ground-based resources, helicopter or aircraft.
 - o Topography, specifically slopes in the terrain. The symbols shows both the direction of the slope and the degree of slope in the form of a scale of weak - medium - steep slope.
 - o Road system, where restrictions with regard to the type of vehicle (light vehicles, light off-road vehicles, etc.) are found but also blocked roads, closed road and one-way road.
 - o Locations in the fire area and nearby areas that affect the development of the fire. This could be, for example, a changed fuel type that results in increased/decreased spread rate or changes in topography (slopes) that result in changed fire behavior.
 - o Sensitive point with regard to WUI (Wildland Urban Interface), marks buildings, facilities, infrastructure, objects worthy of protection etc. where the risk of fire spreading from forests should be taken into account.
 - o Special risks in the form of live power lines where a symbol for a non-live power line is also found. Non-live power lines also include that the power line is protectively grounded. The symbol also indicates safety distances for ground personnel.
 - o Helicopter landing pad with location.
 - o Assembly point with location.
 - o Sector boundary is indicated by a dashed line.
 - o Blocking.
- Wind direction and wind speed: indicated by an arrow in the wind direction and wind speed in m/s and also the wind direction in letters.

Fire parameters • The

direction of spread of the fire front is indicated by a filled arrow.

- Left/right flank and high/low spread rate are indicated with an unfilled arrow of different size depending on the level of spread rate. The purpose is, among other things, to clarify whether any of the flanks has a higher and more critical spread rate.
- The starting point of the fire. Indicated by a filled star. The symbol can also be used to show the starting points of aircraft fires.
- Spread of the fire: solid line with associated time indicates spread of the fire; predicted spread is indicated with dash-dot line. The time information can also advantageously include dates when many forest fires continue for several days.
- Fire band. A solid line symbolizes an active fire band. In the case of an inactive fire band, slanted lines are added along the fire band.
- Active fire areas are indicated by diagonal lines in red. Confirmed burned areas are indicated by diagonal lines in black in the other direction.

Tactics and actions

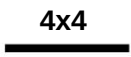
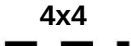
- Water bombing. A circle (helicopter) or oval (airplane) marks an area where water bombing from the air is either underway or planned.
- Evacuation, with a circle - with the letters EV - marks an area where evacuation is either planned, ongoing or has been carried out.
- Protection line (aimed at protecting objects that are deemed to be worth protecting - e.g. buildings - and can be, for example, a mobile hose system or measures taken at buildings to minimize the risk of ignition or stop the spread of fire). The symbol can, for example, be combined with the symbol for a sensitive point (WUI) to indicate that protective measures are planned or implemented in connection with the building, facility, etc.
- Direct attack is drawn with a line and arrows at the end, where the area where direct attack is in progress or planned. This symbol marks which flank the personnel should work towards.
- Limitation line and wet/dry method, with diamond-shaped lines the area where the planned or completed limitation line is found is marked. This symbol is executed where the wet and dry methods are distinguished (if desired) with the letters V and T respectively, alternatively the symbol is drawn in blue for the wet and in red for the dry.
- Ignition is marked with a line and arrow where the arrow indicates the desired direction of spread.

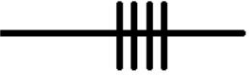
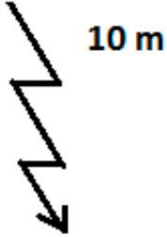
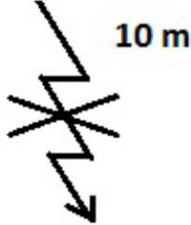
Symbol library

Symbol library customization - Sweden

The symbols in the symbol library below differ, in some cases, compared to the corresponding SiTaC symbols used in the Mediterranean countries. The difference lies partly in the appearance in some cases but also in the fact that the Swedish symbol library has been expanded with additional symbols. A simpler info sheet in Portuguese, Spanish, French, Italian and Croatian will be available during the summer of 2019 to describe the "Swedish" SiTaC library for any incoming resources from southern Europe.

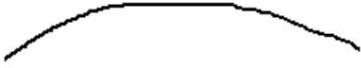
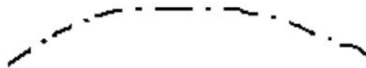
Area of operation

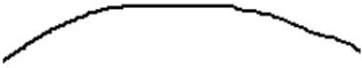
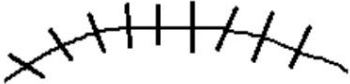
Water intake	
	Water intake for ground-based resources such as tankers, motor sprayers, etc.
	Helicopter
	Aircraft
Slopes (arrow should point in a downward direction)	
	Slight slope
	Medium slope
	Steep slope
Ways	
	Weigh heavy vehicles
	Road heavy off-road vehicles
	Weigh light vehicles
	Road light off-road vehicles

	Road blocked
	Planned road closure
	Closed road
	One-way direction of travel. Half arrowhead indicates direction of travel.
Risk objects and protection objects	
	Natural changes that affect the development of the fire, e.g. changes in vegetation that lead to an increase or decrease in the spread rate or fire barrier.
	Sensitive point with regard to WUI (Wildland Urban Interface), i.e. buildings, infrastructure, objects worthy of protection, etc.
	Live power line, 10 m safety distance
	Voltage-free and protectively grounded power line
Other	
	Helicopter landing pad

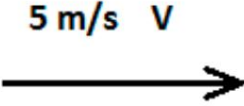
	Assembly point (evacuation)
SEKT 	Sector boundary
	Planned closure
	Barrier

The fire

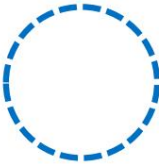
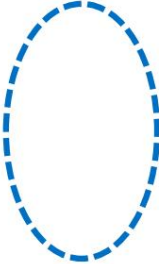
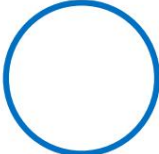
	Direction of spread of the fire front
	The starting point of the fire
	Right flank, high spread rate
	Left flank, low spread rate
14:15 	The spread of the fire with time (the time is given together with the position of the active fire band)
16:15 	Forecasted spread with time indication

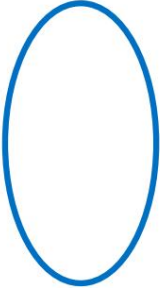
	Active fire band
	Non-active fire band
	Active fire area
	Burned area identified

Wind direction and wind speed

	The arrow indicates the wind direction; the wind speed is given in m/s; the letter indicates where the wind is blowing from (in this case from the west).
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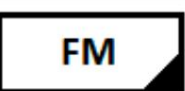
Actions

	Planned or ongoing water bombing with a helicopter. When water bombing is in progress, the symbol is supplemented with the current resource symbol.
	Planned or ongoing water bombing with aircraft. In the event of ongoing water bombing, the symbol is supplemented with the current resource symbol.
	Water bombing carried out by helicopter

	Water bombing carried out by aircraft
	Planned evacuation
	Ongoing evacuation
	Evacuation completed
	Planned or ongoing protection line. If a protection line is in progress, the symbol is supplemented with the current resource symbol.
	Completed protection line
	Planned or ongoing direct attack. In the event of an ongoing direct attack, the symbol is supplemented with the current resource symbol.
	Direct attack carried out
In 	Planned or ongoing restriction line, wet method. In the case of an ongoing restriction line, the symbol is supplemented with the current resource symbol.
T 	Planned or ongoing restriction line, dry method. In the case of an ongoing restriction line, the symbol is supplemented with the current resource symbol.
In 	Completed boundary line, wet method

T 	Completed boundary line, dry method
	Planned or ongoing ignition on line. When ignition is in progress, the symbol is supplemented with the current resource symbol.
	Completed ignition on line

Resources

	Planned emergency service resource
	Emergency services resource on site
	Planned ambulance resource
	Ambulance resources on site
	Planned police resource
	Police resources on site
	Planned other resource (the organization is indicated by abbreviation or in plain text in the box).
	Other resources on site
	Planned flying unit
	Flying unit on site
	Planned management location

	Existing management site
	Planned breakpoint
	Existing breakpoint



Myndigheten för
samhällsskydd
och beredskap